SOFTWARE DESIGN DOCUMENT

NINAAllskySafetyMonitor

NINA safety-monitor plugin using Allsky camera image analysis and WunderStation ISYDNEY478 weather data

Document	Software Design Document
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Status	Draft for review (slices 1–5 implemented and verified; slice 6 implemented, awaiting verification)

Revision history

Version	Date	Change
1.0	1 Nov 2026	First issue: use case model, class design, scenarios, traceability matrix.
1.1	1 Nov 2026	Updated to match the slice 1 code: MonitorHost and singleton settings store, IMonitoringCycle and MonitorPorts, settings read every cycle, views as XAML resource dictionaries, source-scanning coverage tests and test fakes, extra settings fields. REQ-06 rationale reworded. See §1.10.
1.2	2 Nov 2026	Slices 2–4 as built (acquisition, weather, decision, override, decision log, redaction) and requirements v5 working area (REQ-22, REQ-25, REQ-26, REQ-58). See §1.9.
1.3	2 Nov 2026	Slice 5a as built (mount, coordinates, working area, ROI, slew hold, mount watcher) and the slice 5b design (lens calibration from catalogue stars, working-area outline, enlarged image viewer); requirements v5.2 (REQ-20 fit limits, REQ-27 mount interval, REQ-59). See §1.8.
1.4	2 Nov 2026	Slice 5b as built and verified: frame decoding behind IFrameDecoder (WPF), darkest-channel star detection ranked by peak, the calibration helper classes, overlay content, viewer close handling, slew moves ignored by the mount watcher; generic Imager actor. See §1.7.
1.5	2 Nov 2026	Slice 6 design (cloud detection) for requirements v5.3: limiting magnitude 4.4, exposure-independent brightness test against a clear-sky map, sky-frame check (REQ-60), tuning recorder saving every frame with 30-day retention, Moon calculator, status metrics and history. See §1.6.

Version	Date	Change
1.6	3 Nov 2026	Slice 6 as built: CloudDetector returns the captured reference for the host to save, the tuning recorder runs after each decision, settings migration, image size recorded with the lens, block map and metric details. See §1.5.

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1 Introduction

1.1 Purpose

This document describes the design of NINAAllskySafetyMonitor, a NINA plugin that reports whether it is safe to image. It analyses the latest image from the roof-mounted Allsky camera for cloud in the region of sky the telescope is pointing at, combines that with rain and wind data from weather station ISYDNEY478, and reports the result to NINA as a safety monitor device.

1.2 Scope

The design covers the use case model, the object-oriented class structure, the main and failure scenarios, and traceability from every requirement in requirements.md v5.3 to the design elements that realise it. Implementation details below class level are left to the code.

1.3 References

Ref	Document
[R1]	NINAAllskySafetyMonitor Requirements, version 5.3, 2 November 2026 (requirements.md)
[R2]	N.I.N.A. source code, github.com/isbeorn/nina (commit 717ea13, 23 Oct 2026): IEquipmentProvider, ISafetyMonitor, ITelescopeMediator, NINA.Astrometry
[R3]	N.I.N.A. plugin template, github.com/isbeorn/nina.plugin.template
[R4]	Weather.com PWS current-observations API (response format confirmed 1 Nov 2026)
[R5]	Yale Bright Star Catalogue, 5th revised edition (Hoffleit and Warren, 1991; CDS V/50), public domain

1.4 Conventions

- Diagrams use UML notation drawn in the Rational Rose style. Sources are PlantUML text files kept with the code.
- The dotted label at the foot of each class (for example `REQ-23 REQ-26`) lists the requirements the class implements. In code these become `[Requirement("REQ-23", "REQ-26")]` attributes.
- Defaults shown in brackets in the requirements are held in the JSON settings (§2.5); class diagrams show them as initial values.

1.5 Changes in version 1.6

Version 1.6 brings the design into line with the slice 6 code (375 unit tests pass; NINA verification to follow). The behaviour designed in 1.5 is unchanged; these are the differences found while building it.

Diagram	Change	Reason
CL-07, SD-03	CloudDetector.Assess returns CloudDetection: the assessment plus any clear-sky reference captured from the frame. The cycle uses it at once and hands it to MonitorPorts.ReferenceCaptured, which the plugin sets to PluginSettingsStore.SaveReference.	Core does not write the settings file; the host saves, as for calibration (REQ-57).
CL-07	Reference capture runs the star test over the whole working area only when no reference exists, the thresholds are tuned, the Moon is down and the ROI is not cloudy. ClearSkyReferenceBuilder.CanCapture added.	The whole-area search takes about 2.5 s on a 2500 px frame; a normal cycle about 0.4 s.
CL-07	RoiMaskBuilder keeps the working-area mask until the lens, area or image size changes; ExpectedStarPredictor takes the ROI mask; StarMatcher returns one entry per expected star; StarDetector searches only the ROI bounds.	Speed, and one place that decides which pixels are in the ROI.
CL-07	BackgroundStatistics uses percentiles (non-uniformity = (P84 − P16) ÷ 2 ÷ median) instead of removing stars first. BlockStatistics stores the frame size and NaN for unmeasured blocks, written to JSON as one line of numbers per array (CompactFloatArrayConverter).	Robust without a star list; the map adds a few lines to settings.json, not thousands.

Diagram	Change	Reason
CL-02, CL-06	MoonState record (position, illumination); MoonCalculator.State; CloudAssessment gains Summary and Moon; PixelRect and PixelMask.Bounds. LensCalibration records ImageWidth and ImageHeight.	The reason text and the Moon line come with the assessment; the sky-frame size check needs the calibrated size (0 for older or hand-entered lenses, check skipped).
CL-08, CL-09, SD-02	TuningRecorder subscribes to SnapshotAvailable, like DecisionLogger, instead of being called inside the cycle; folders are per night (local date 12 h earlier). The cycle decodes once per image and runs the analysis off the caller's thread. StatusHistory removed: the chart view model keeps tonight's points.	Rows carry the decided state; a night is not split at midnight; one less class.
CL-03b	SettingsMigrator: SchemaVersion 2 moves a saved default limiting magnitude 4.0 to 4.4.	Saved files keep old defaults otherwise.
CL-09	DecisionLogger gains stars expected and found, star fraction, relative brightness, non-uniformity ratio and the Moon.	REQ-44: the metrics behind each decision.
CL-10a, CL-10b, SD-11	MetricsViewModel lines (stars, brightness, Moon, reference); HistoryChartViewModel series; SettingsViewModel cloud and tuning fields, reference status and Clear button, and Save keeps the saved reference; OverlayContent.CloudStars; viewer Expected stars toggle.	REQ-45, REQ-46; a reference captured while the Options page is open is not lost.
CL-11	Fakes: FakeFrameDecoder, SyntheticCloudSky (1000 px frames with cloud patches, exposure changes and a notice image).	Cloud detection tested end to end without real frames.

1.6 Changes in version 1.5

Version 1.5 designs slice 6 (cloud detection) against requirements v5.3. A test on the clear 20 October 2026 frame with the calibrated lens found every expected star in 15° ROIs in open sky, but at magnitude 4.0 many ROIs expected fewer than 8 stars, and ROIs reaching behind the trees found only about 40%. Hence the 4.4 limit and the rule that only stars inside the working area and its outline are expected.

Diagram	Change	Reason
CL-07, SD-03	Cloud detection designed in full: RoiMaskBuilder, ExpectedStarPredictor (ROI ∩ working area ∩ outline, away from the Moon), StarMatcher (tolerance from the calibration RMS), StarFractionTest, BackgroundStatistics on 32 px blocks, BrightnessTest, ClearSkyReferenceBuilder, CloudDetector.	REQ-28 to REQ-31 v5.3.
CL-07, CL-03b	Brightness test made exposure-independent: per-block ratio to a clear-sky map, ROI relative to the rest of the working area; non-uniformity ratio is scale-free. ClearSkyReference becomes a map (BlockStatistics). LimitingMagnitude 4.4.	Allsky auto-exposure keeps overall brightness steady; the western sky is far brighter than the east (REQ-30 v5.3).
CL-07, SD-03, UC-Ca	SkyFrameCheck: wrong size, or a flat background inside the working area, is "Not a sky frame" (a failed NoCloud check).	REQ-60: Allsky shows a drawn notice image when its software stops.
CL-06	MoonCalculator (position and illumination) in Core; NovasAstroCalculator delegates to it.	REQ-31; testable without NINA, like CoordinateConverter.
CL-08, SD-02, SD-01	MonitoringCycle decodes the frame through IFrameDecoder (MonitorPorts.FrameDecoder = WpfFrameDecoder) and runs the sky-frame check, cloud detection and tuning recorder.	The placeholder "Cloud detection not implemented yet" goes.
CL-09, CL-03b	TuningRecorder saves every frame and a CSV row per cycle in daily folders; deletes folders older than RetentionDays [30].	REQ-41 v5.3.
CL-02, CL-10b	CloudAssessment carries relative brightness, the non-uniformity ratio and the expected/found star positions; status panel metrics and the night's history graph (OxyPlot); expected stars drawn green (found) or red (missing).	REQ-45.

1.7 Changes in version 1.4

Version 1.4 records slice 5b as built and verified (all 18 checks passed on 2 November 2026; 51 of 59 requirements verified). On the 20 October 2026 frame the automatic calibration matched 227 stars with an RMS error of 2.25 px. The behaviour and requirement allocation are unchanged from 1.3.

Diagram	Change	Reason
CL-03a, CL-07, CL-10b	GrayFrameDecoder in Core replaced by the IFrameDecoder port, implemented by WpfFrameDecoder in the plugin; JpegBitmap loads images for display.	Keeps Core free of an imaging library (ImageSharp not needed).
CL-02, CL-07	StarDetector works on the darkest of R, G, B per pixel and sorts spots by peak; DetectedStar gains Peak and AreaPx.	The camera's noisy pixels are coloured and stars are white: 28 of the 30 brightest spots on the test frame are stars.
CL-09	Calibration helpers as built: SkyStar and SkyStars (catalogue positions at the frame time), SpotIndex (grid search), StarMatch, LevenbergMarquardt (own solver).	Design decision 2 Nov 2026: own solver instead of MathNet.Numerics.
CL-10b, SD-08, SD-12	OverlayContent record for the overlay; CalibrationMessage shown under the calibration buttons; the viewer opens after the click finishes and closes once, deferred, ignoring focus loss while closing; every viewer handler catches its own errors.	A second Close during closing crashed NINA (slice 5b check 12); the message was out of sight at the foot of the page (check 14).
CL-08	MountWatcher ignores position changes while the mount is slewing.	Only the start and end of a slew matter; the log had a line every 2 s during slews.
UC-01, UC-02	The Imager actor is generic.	The requirements no longer name a person.

1.8 Changes in version 1.3

Version 1.3 brings the design into line with slice 5a as built and verified, and designs slice 5b. A prototype fit on a clear frame from 20 October 2026 (allsky-20261020022133.jpg) matched 33 catalogue stars with an RMS error of 2.7 px (0.15°), confirming the calibration approach; the zenith is 176 px left of the image centre and north is rotated −5.6°.

Diagram	Change	Reason
CL-06	As built: CoordinateConverter (IAU GMST, J2000 precession, about 0.01° against astropy), SkyGeometry.Destination, LensProjection.Approximate and FromHorizonRadius, MountStateInterpreter.WhyUnusable, RoiCalculator returns the reason, RoiBuilder judges the circle on the sky from 400 sample points. LensCalibration gains StarsMatched, RmsErrorDeg and Manual.	RA/Dec ↔ alt/az moved from NINA.Astrometry to Core so it is unit-tested; NovasAstroCalculator delegates to it. The ROI does not depend on the lens model.
CL-08, SD-01, SD-05	MountWatcher reads the mount every 2 s and asks MonitorService to re-evaluate at once on a slew, park, tracking or position change; the cycle fetches the image at most once per image interval. SafetyDecisionEngine holds the last decision while slewing.	REQ-27 v5.2: a slew is held within seconds whatever the image interval (slice 5a check 15).
CL-10a	NinaMountStateSource pulls TelescopeInfo from ITelescopeMediator (imported by the provider) instead of registering as a consumer.	Simpler; the watcher and the cycle ask when they need it.
CL-02, CL-03b	MonitorSnapshot carries the MountState; RegionOfInterest centre is optional and has a note. MountSettings (PollIntervalS 2) and LensFitSettings (MinMatchedStars 15, MaxFitErrorDeg 0.2°, catalogue mag 4.5).	REQ-27 and REQ-20 v5.2.
CL-07, CL-09, SD-08	Slice 5b: StarCatalogue embedded from the Yale Bright Star Catalogue; StarDetector removes hot pixels; Calibration package redesigned: FrameTimeReader, InitialGuessSearch, StarIdentification fallback, LensFitter, CalibrationQualityCheck, LensCalibrationSession.	REQ-19 to REQ-21 and the fit-limit warning (REQ-20). Frame time from the archive file name or the live image capture time.
CL-10b, SD-12	Slice 5b: ImageViewerWindow with View mode (from the status panel; outside click or Esc closes) and Calibration mode (Save or Cancel only); cursor alt/az; overlay layers. SettingsViewModel gains calibration commands; CalibrationTemplates replaced by the viewer.	REQ-59 v5.2; the panel image is too small to click stars or draw an outline.
CL-09	DecisionLogger columns for mount state, RA/Dec, reported Alt/Az, ROI alt/az, mode, % inside and note; a file started by an older version gets the current header.	REQ-06 diagnostics; slice 5a check 11.

1.9 Changes in version 1.2

Version 1.2 brings the design into line with the code built and verified in slices 2 to 4, and designs the working area added in requirements v5 (to be built in slice 5).

Diagram	Change	Reason
CL-02, CL-03b, CL-06, SD-05	Horizon mask replaced by the working area: WorkingAreaDefinition (centre alt/az, radius, outline) and WorkingArea. RoiBuilder clips the 15° circle to the working area and uses the whole area when the mount position is unavailable or less than MinInsideFraction is inside. RegionOfInterest gains UsesWholeWorkingArea, InsideFraction and Note.	Requirements v5: REQ-22, REQ-25, REQ-26, REQ-58.
CL-03a, CL-03b	CredentialKeys; DirectSourceSettings.HostKeyAlgorithm (ecdsa-sha2-nistp256); web file name allsky.jpg; RoiSettings.MinInsideFraction.	The pinned fingerprint is the Pi's ECDSA key; image.jpg on the web server is not the current image (slice 2 checks 2 and 10).
CL-04, SD-06	IImageSource gains force and Reset; FetchResult.Untrusted; IRemoteFileConnector/Session separate SSH.NET from the logic; ImageSourceFactory.For; ImageFetchTester. SSH.NET 2026.0.0.	Testable without a Pi; a host-key mismatch discards earlier images (REQ-52); advisories against 2023.0.1.
CL-05	WeatherFetchResult carries the error; WeatherPoller is asked every cycle and polls when due; WeatherFetchTester; the client never puts the API key in messages.	Readable reasons; REQ-32, REQ-54.
CL-08, SD-02, SD-04, SD-10	The decision engine and manual override run in MonitorService on every snapshot, including failed ones; SafetyConditionsBuilder removed (the cycle gathers the conditions); image and weather fetched in parallel; override held by MonitorHost and applied at once.	A failed cycle restarts the safe-after period; override effective immediately (REQ-38 to REQ-40).
CL-09	DecisionLogger writes a daily CSV (Excel) and deletes old files; RedactingLogSink replaces RollingFileLogSink; SecretRedactor reads the current secrets each time.	REQ-44, REQ-54; NINA keeps its own text log.
CL-10a, CL-10b, SD-11	WindowsCredentialStore over advapi32; PasswordBoxAssistant; Test image and Test weather; status panel image, weather line, countdown and override banner.	REQ-45, REQ-46, REQ-53.
CL-11	Fakes for image sources, SFTP, HTTP, weather and credentials.	Core testable without network.

1.10 Changes in version 1.1

Version 1.1 brings the design into line with the code written for slice 1 (skeleton) and with decisions taken while writing it. The behaviour and requirement allocation are unchanged.

Diagram	Change	Reason
CL-02	HorizontalCoordinate and PixelPoint are record structs.	Small values created in large numbers during star matching; structs avoid heap allocation.
CL-03b	Added SchemaVersion, DirectSourceSettings.Port, WeatherSettings.TimeoutS; SettingsStore gains Path, ToJson, FromJson, Clone; write-then-replace save.	Settings migration, non-standard SSH port, weather timeout (REQ-51), safe editing of a copy and crash-safe saving.
CL-06	SiteLocationResolver takes the fallback as a function; a profile at 0°, 0° counts as unset. OperatingWindow takes the limit per call.	Settings are read live each cycle; NINA profiles default to 0°, 0°.
CL-08	Added IMonitoringCycle and MonitorPorts; MonitorService takes interval and FailSafeGuard takes timeout as functions; MonitoringCycle reads settings each cycle.	Testable with a fake cycle; one object carries the host’s adapters; saved settings apply without reconnecting.
CL-10a	Added MonitorHost (single instance). PluginSettingsStore is a single instance holding the current settings. The device delegates to MonitorHost.	NINA creates the plugin, device and dock panel separately; they must share one monitor and one settings copy.

Diagram	Change	Reason
CL-10b	Views are XAML resource dictionaries (StatusTemplates, OptionsTemplates, CalibrationTemplates) matched by key. SettingsViewModel edits a copy, with Save and Revert. Added DelegateCommand. Duplicate Views package removed.	How NINA locates plugin views; edits apply only on Save.
CL-11	Coverage tests scan source files; added RequirementTagScanner, RepositoryPaths, RequirementInfo, CoverageReport and the Fakes package.	The plugin assemblies cannot be loaded on the Mac; fakes make Core testable without NINA.
SD-01, SD-07, SD-10, SD-11	Connect, failure, disconnect and status/settings flows go through MonitorHost; settings read every cycle; Revert added.	Follows from CL-08 and CL-10a/b.
§2.3, CL-06	REQ-06 rationale reworded to match requirements.md.	Research on the AM5N home-referenced Azimuth frame.

2 Architecture

2.1 Principles

The design is object-oriented with single-role, single-responsibility classes, one class per file. All safety logic is in a class library with no NINA references, so every rule can be unit-tested and replayed offline. NINA-specific code is confined to a thin plugin layer that adapts NINA to the library's interfaces (ports).

2.2 Assemblies

Assembly	Role	Depends on
NINAAllskySafetyMonitor.Core	All safety logic: acquisition, weather, geometry, cloud detection, decision, monitoring loop, tuning, logging, settings, traceability attribute.	— (no NINA)
NINAAllskySafetyMonitor.Astrometry	Implements IAstroCalculator over NINA.Astrometry (NOVAS): sun and Moon position, RA/Dec ↔ alt/az.	Core, NINA.Astrometry
NINAAllskySafetyMonitor.Plugin	NINA plugin: equipment provider, safety-monitor device, adapters for mount, site, credentials and logging, MVVM views.	Core, Astrometry, NINA
NINAAllskySafetyMonitor.Harness	Console tool: offline replay of saved frames (REQ-42) and generation of the traceability workbook.	Core, Astrometry
NINAAllskySafetyMonitor.Tests	Unit tests per class and the requirement-coverage tests.	Core

2.3 NINA integration

Confirmed against the NINA source [R2]: the plugin exports an `IEquipmentProvider<ISafetyMonitor>` through MEF. NINA gathers it with `PluginEquipmentProviders<ISafetyMonitor>` and lists the returned device in Equipment → Safety Monitor alongside other drivers. The device implements `ISafetyMonitor` (`IsSafe` plus the `IDevice` members). Connect builds and starts the monitor; Disconnect stops it, so nothing runs outside NINA (REQ-01, REQ-02).

Mount state is read from `ITelescopeMediator.GetInfo` (imported by the equipment provider) every 2 s by MountWatcher and once per cycle; site location comes from `IProfileService`. The ROI is always computed from RA/Dec. The driver's reported Alt/Az is logged for diagnostics only, because the ASI Mount driver reports Azimuth in the mount's own home-referenced frame (0° at the celestial pole), not the geographic azimuth the ASCOM standard defines (REQ-06).

NINA creates the plugin, the equipment provider (and its device) and the dock panel independently. They share one `MonitorHost` and one `PluginSettingsStore` (single instances), so there is one monitor and one copy of the settings per NINA session. Connect calls `MonitorHost.StartAsync`; Disconnect and plugin teardown call `StopAsync` (CL-10a, SD-01, SD-10).

2.4 Threading and fail-safe behaviour

MonitorService runs the monitoring cycle on a background task, never on NINA's UI thread. Within a cycle the image fetch and the weather poll run in parallel, each with its own timeout. Every snapshot, including one from a failed cycle, then passes through SafetyDecisionEngine (unsafe at once, safe after the clear period) and ManualOverride before it is published. Each cycle runs inside FailSafeGuard, which applies a timeout and a catch-all; any exception, hang or undeterminable state yields an Unsafe status and is logged, and nothing is thrown back to NINA (REQ-49, REQ-50). Every network operation has its own timeout (REQ-51). View models marshal snapshots to the UI dispatcher.

2.5 Settings

All configurable values live in one JSON file (the MonitorSettings tree, CL-03b), loaded and saved by SettingsStore, including the lens calibration, working area and clear-sky reference (REQ-55 to REQ-57). The SFTP password, if used, is held in Windows Credential Manager, and secrets are redacted from all logs (REQ-53, REQ-54). The settings file is `%LOCALAPPDATA%\NINAAllskySafetyMonitor\settings.json`,

outside the NINA plugin folder, so reinstalling the plugin keeps it. Saves go to a temporary file which then replaces the old one. MonitoringCycle reads the current settings at the start of every cycle, so a connected monitor applies saved changes on its next cycle without reconnecting.

2.6 Traceability process

Every class and every test carries `[Requirement("REQ-xx")]`. A coverage test parses requirements.md, scans the source files for tags (so the Windows-only projects are covered when tests run on the Mac), and fails the build when a requirement has no implementing class or no test, or when a tag names an unknown requirement. The two completeness checks are skipped until the last slice; a coverage report lists the gaps meanwhile. The harness regenerates traceability.xlsx for review. Section 6 gives the design-level matrix.

2.7 Candidate libraries

Need	Candidate	Notes
SFTP with key login and host-key pinning	SSH.NET 2026.0.0 (in use)	With BouncyCastle.Cryptography and Microsoft.Extensions.Logging.Abstractions; host-key type restricted to the pinned key
JPEG decode and pixel access	WPF imaging (in use)	Behind IFrameDecoder in the plugin; no extra library or licence
Lens-model least-squares fit	Own LevenbergMarquardt (in use)	6 parameters (centre, rotation, c1–c3), numerical Jacobian; no extra library
History chart	OxyPlot	Shipped with NINA; referenced, not copied
Star catalogue	Yale Bright Star Catalogue [R5]	Stars brighter than mag 4.5 (about 900), embedded in the Core DLL as Resources/BrightStars.csv, names for the brightest

SSH.NET confirmed in slice 2; WPF decoding, the own solver and the catalogue in slice 5b; OxyPlot to be confirmed in slice 6.

3 Use case model

Six actors interact with the monitor: you as the imager, the NINA sequencer, a timer that drives the monitoring cycle, and three external systems (the Allsky Pi or web server, the Weather.com PWS API, and the mount via NINA). UC-A covers operation, UC-B covers set-up and tuning, and UC-Ca/Cb trace each requirement group to a use case.

UC	Use case	Actors	Requirements
UC-01	Connect safety monitor	Imager, NINA	REQ-01–02, REQ-48
UC-02	Disconnect safety monitor	Imager, NINA	REQ-01–02
UC-03	Report safety status	NINA	REQ-03
UC-04	Run monitoring cycle	Timer	
UC-05	Determine operating window	—	REQ-04, REQ-08–10
UC-06	Acquire Allsky image	Allsky Pi / web server	REQ-11–18, REQ-52–53, REQ-60
UC-07	Determine region of interest	Mount (via NINA)	REQ-04–07, REQ-23–27, REQ-58
UC-08	Detect cloud in ROI	—	REQ-28–31
UC-09	Assess weather	Weather.com PWS API	REQ-32–36
UC-10	Decide safety	—	REQ-37–39
UC-11	Override safety manually	Imager	REQ-40
UC-12	Calibrate lens	Imager	REQ-19–21, REQ-47
UC-13	Define working area	Imager	REQ-22, REQ-47
UC-14	View status	Imager	REQ-45, REQ-59
UC-15	Edit settings	Imager	REQ-46, REQ-53, REQ-55–57
UC-16	Record tuning data	Timer	REQ-41, REQ-43
UC-17	Replay frames offline	Imager	REQ-42
UC-18	Log decision	—	REQ-44, REQ-54
UC-19	Handle monitor failure	—	REQ-49–51

Figure 3-1 UC-A Operation — monitoring and reporting to NINA

The timer drives UC-04, which includes the operating window, image acquisition, ROI, cloud detection, weather and decision use cases. UC-10 includes reporting to NINA and logging. Manual override and failure handling extend the flow.

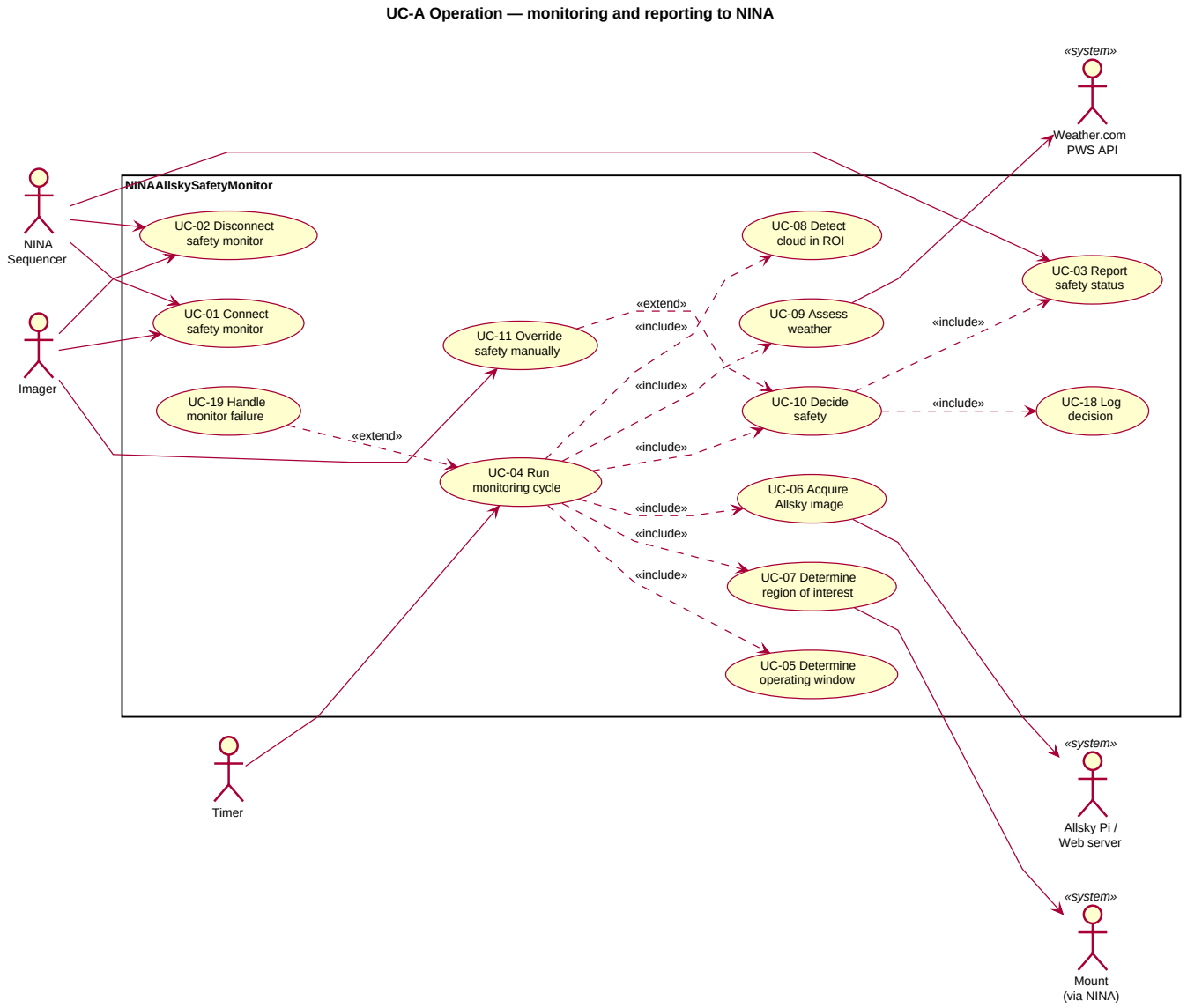


Figure 3-2 UC-B Set-up, review and tuning

Calibrating the lens and defining the working area both use a live image. Tuning-data recording extends the monitoring cycle when tuning mode is on; offline replay reuses cloud detection.

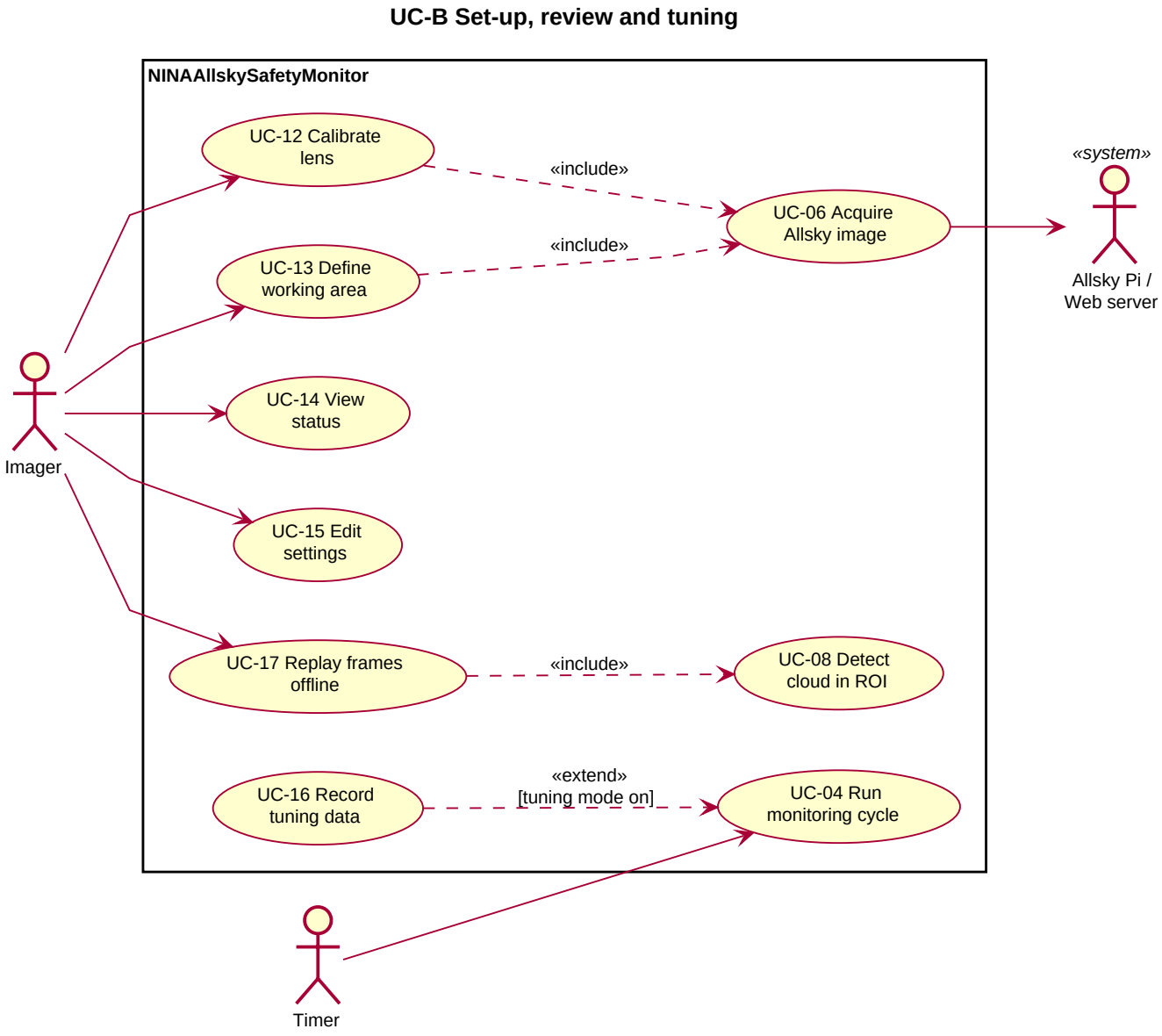


Figure 3-3 UC-Ca Requirements §1–§8 → use cases

Functional requirement groups for NINA integration, site and mount, operating window, acquisition, calibration, ROI, cloud detection and weather.

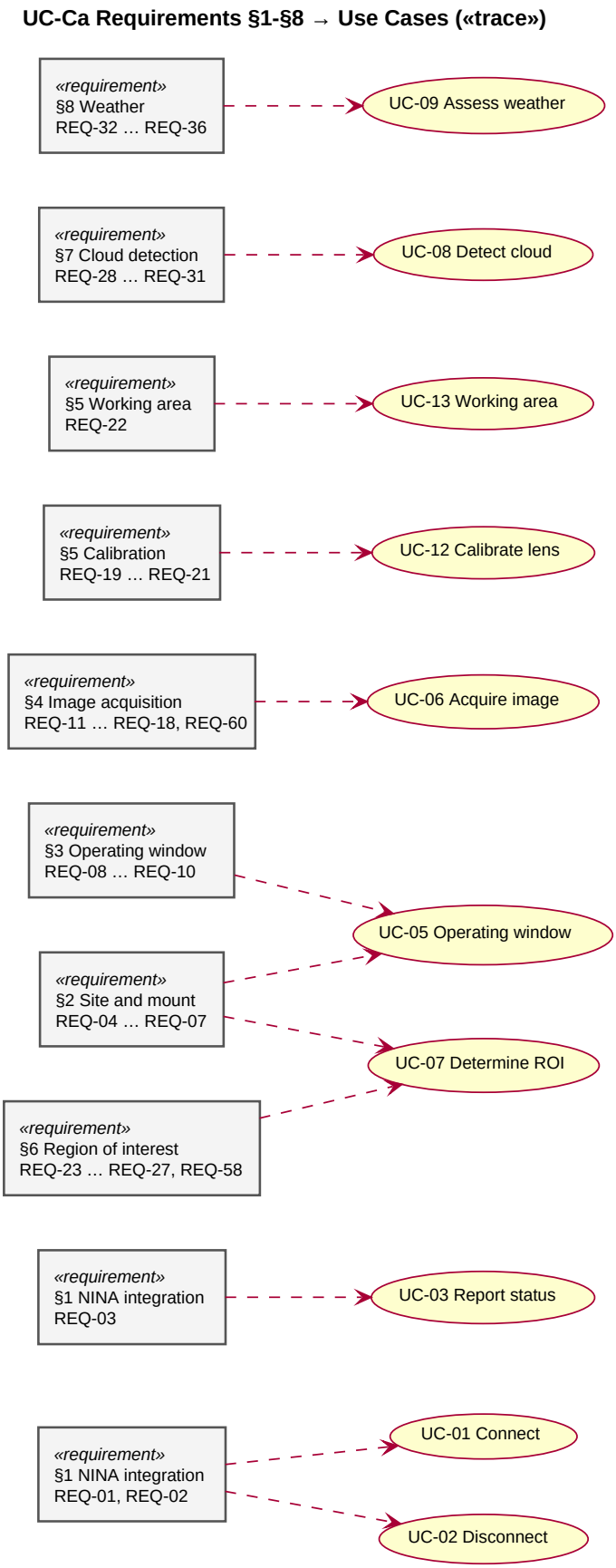
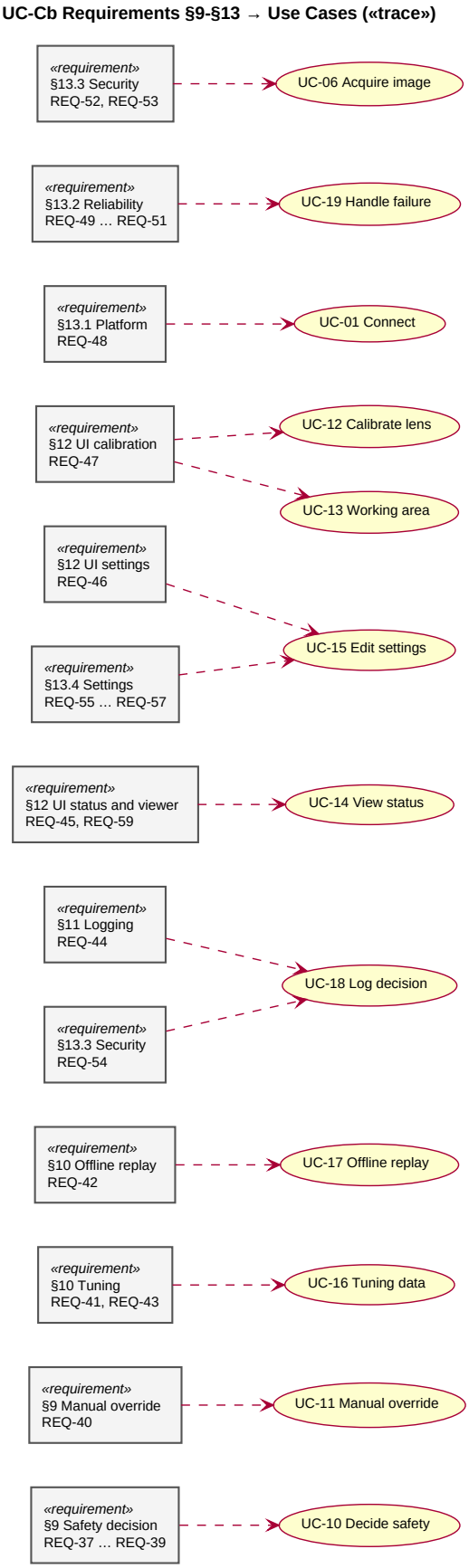


Figure 3-4 UC-Cb Requirements §9–§13 → use cases

Safety decision, tuning, logging, user interface and the non-functional groups.



4 Class design

Thirteen class diagrams describe the design package by package. Table 4-1 lists every class that implements a requirement, with its diagram.

Table 4-1 Classes and the requirements they implement

Diagram	Class	Requirements
CL02	CloudAssessment	REQ-28, REQ-45
CL02	MonitorSnapshot	REQ-41, REQ-45
CL02	MoonState	REQ-31
CL02	MountState	REQ-05–07
CL02	RegionOfInterest	REQ-23, REQ-25–26, REQ-58
CL02	SafetyConditions	REQ-37
CL02	SafetyStatus	REQ-03, REQ-39–40, REQ-43
CL02	SiteLocation	REQ-04
CL02	SkyImage	REQ-11, REQ-15
CL02	WeatherObservation	REQ-33
CL03a	CredentialKeys	REQ-53
CL03a	IAstroCalculator	REQ-08, REQ-24, REQ-31
CL03a	ICredentialStore	REQ-53
CL03a	IFrameDecoder	REQ-20
CL03a	IMountStateSource	REQ-05
CL03a	ISiteLocationSource	REQ-04
CL03b	CalibrationData	REQ-57
CL03b	CloudDetectionSettings	REQ-29–31, REQ-43
CL03b	LensFitSettings	REQ-20
CL03b	MonitorSettings	REQ-55
CL03b	MountSettings	REQ-27
CL03b	RoiSettings	REQ-23, REQ-58
CL03b	SettingsMigrator	REQ-29, REQ-55
CL03b	SettingsStore	REQ-55–56
CL03b	SettingsValidator	REQ-22, REQ-27, REQ-46, REQ-58
CL03b	TuningSettings	REQ-41
CL04	FetchResult	REQ-16, REQ-52
CL04	HostKeyMismatchException	REQ-52
CL04	HostKeyVerifier	REQ-52
CL04	IImageSource	REQ-11
CL04	ImageAcquirer	REQ-14, REQ-17–18
CL04	ImageFetchTester	REQ-46
CL04	ImageFreshnessEvaluator	REQ-15, REQ-18, REQ-50
CL04	ImageSourceFactory	REQ-11
CL04	JpegIntegrityChecker	REQ-17
CL04	SftpConnector	REQ-12–13, REQ-51–53
CL04	SftpFileSession	REQ-11
CL04	SftpImageSource	REQ-11, REQ-15–16, REQ-51
CL04	WebImageSource	REQ-11, REQ-15, REQ-51

Diagram	Class	Requirements
CL05	IWeatherProvider	REQ-32
CL05	PwsObservationParser	REQ-33
CL05	PwsWeatherClient	REQ-32, REQ-51, REQ-54
CL05	WeatherEvaluator	REQ-34–36, REQ-50
CL05	WeatherFetchTester	REQ-46
CL05	WeatherPoller	REQ-32
CL06	CoordinateConverter	REQ-24
CL06	LensCalibration	REQ-19, REQ-60
CL06	LensProjection	REQ-19, REQ-24
CL06	MoonCalculator	REQ-31
CL06	MountStateInterpreter	REQ-07, REQ-26
CL06	OperatingWindow	REQ-08–10
CL06	RoiBuilder	REQ-23, REQ-25–26, REQ-58
CL06	RoiCalculator	REQ-06, REQ-23–24, REQ-26
CL06	SiteLocationResolver	REQ-04
CL06	WorkingArea	REQ-22, REQ-25
CL06	WorkingAreaDefinition	REQ-22, REQ-57
CL07	BackgroundStatistics	REQ-30
CL07	BrightnessTest	REQ-30–31
CL07	ClearSkyReference	REQ-30, REQ-57
CL07	ClearSkyReferenceBuilder	REQ-30
CL07	CloudDetection	REQ-28, REQ-30
CL07	CloudDetector	REQ-28
CL07	ExpectedStarPredictor	REQ-29, REQ-31
CL07	MoonBrightnessAdjuster	REQ-31
CL07	RoiMaskBuilder	REQ-25, REQ-29
CL07	SkyFrameCheck	REQ-60
CL07	StarCatalogue	REQ-20, REQ-29
CL07	StarDetector	REQ-20, REQ-29
CL07	StarFractionTest	REQ-29
CL07	StarMatcher	REQ-29
CL08	FailSafeGuard	REQ-49–51
CL08	ManualOverride	REQ-40
CL08	MonitorService	REQ-02, REQ-27, REQ-38, REQ-40, REQ-49
CL08	MonitoringCycle	REQ-05, REQ-10, REQ-14, REQ-28, REQ-37, REQ-60
CL08	MountWatcher	REQ-05, REQ-27
CL08	SafetyDecisionEngine	REQ-03, REQ-27, REQ-37–39
CL08	SafetyHysteresis	REQ-38–39
CL08	SlewHoldPolicy	REQ-27
CL09	CalibrationQualityCheck	REQ-20
CL09	DecisionLogger	REQ-06, REQ-44, REQ-54
CL09	FrameTimeReader	REQ-20
CL09	InitialGuessSearch	REQ-20
CL09	LensCalibrationSession	REQ-19–20

Diagram	Class	Requirements
CL09	LensFitter	REQ-19–20
CL09	MetricsCsvWriter	REQ-41
CL09	RedactingLogSink	REQ-54
CL09	ReplayImageSource	REQ-42
CL09	SecretRedactor	REQ-54
CL09	SkyStars	REQ-20, REQ-24
CL09	TuningRecorder	REQ-41
CL09	TuningStateEvaluator	REQ-43
CL10a	AllskySafetyMonitorDevice	REQ-01–03, REQ-49
CL10a	AllskySafetyMonitorProvider	REQ-01, REQ-05
CL10a	AllskySafetyPlugin	REQ-46, REQ-48
CL10a	MonitorHost	REQ-02–03, REQ-05
CL10a	NinaMountStateSource	REQ-05–07
CL10a	NinaSiteLocationSource	REQ-04
CL10a	NovasAstroCalculator	REQ-08, REQ-24, REQ-31
CL10a	PluginSettingsStore	REQ-30, REQ-55–57
CL10a	WindowsCredentialStore	REQ-53
CL10b	CalibrationViewModel	REQ-19–22, REQ-47, REQ-57
CL10b	HistoryChartViewModel	REQ-45
CL10b	ImageViewerViewModel	REQ-21, REQ-59
CL10b	MetricsViewModel	REQ-45
CL10b	OverlayImageComposer	REQ-21–23, REQ-45
CL10b	SettingsViewModel	REQ-22–23, REQ-27, REQ-46, REQ-53, REQ-55, REQ-58
CL10b	StatusViewModel	REQ-40, REQ-43, REQ-45, REQ-58–59
CL10b	WpfFrameDecoder	REQ-20
CL11	FixedMountStateSource	REQ-42
CL11	ReplayClock	REQ-42
CL11	ReplayCommand	REQ-42

Figure 4-1 CL-01 Assemblies and packages

Core holds all safety logic and has no NINA references. Domain types are used by every package; those dependencies are omitted for clarity.

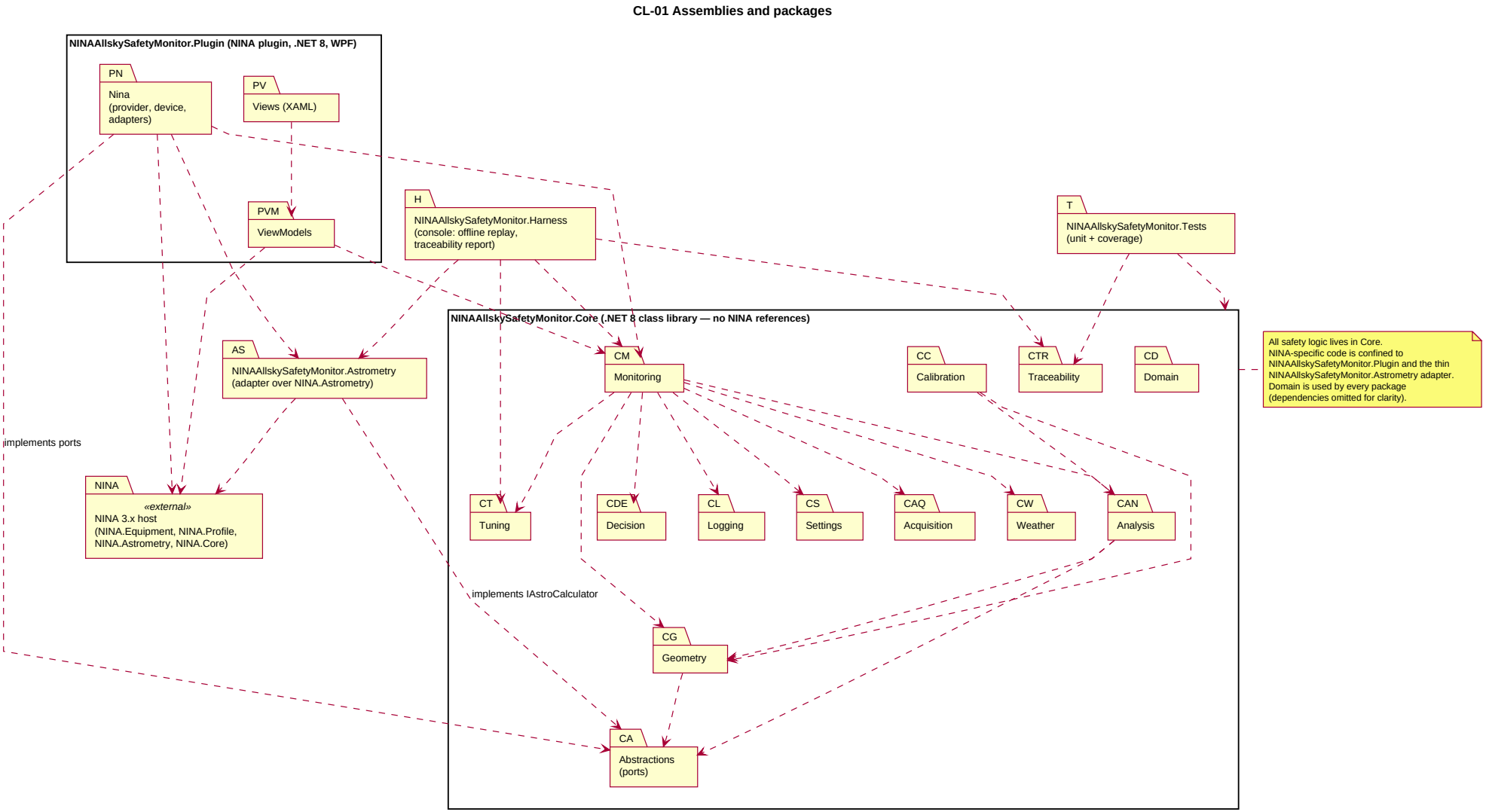


Figure 4-2 CL-02 Domain

Immutable value objects passed between components: images, mount state, ROI, assessments, conditions, safety status and the snapshot used by the UI and tuning.

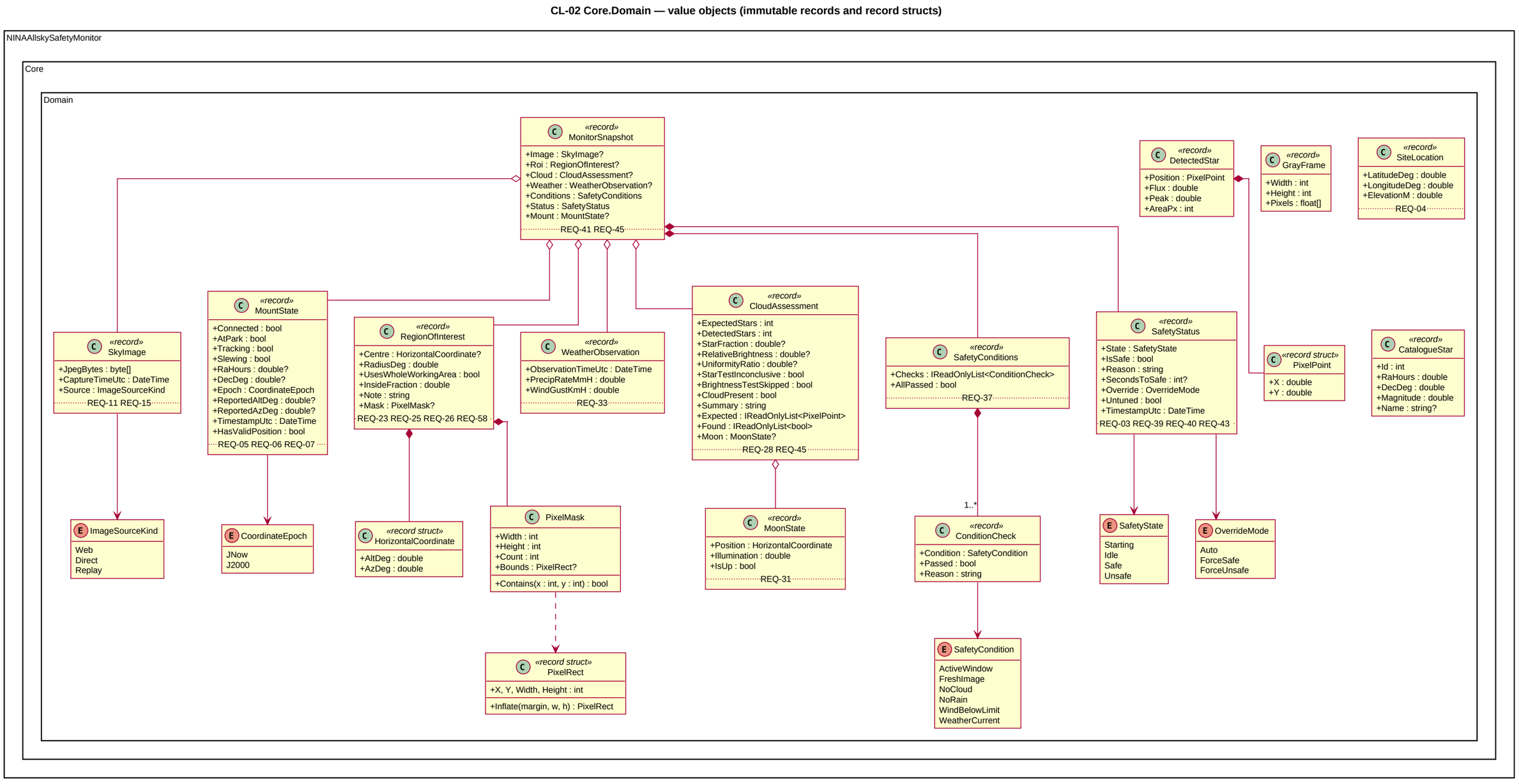


Figure 4-3 CL-03a Ports

Interfaces Core needs from its host. The plugin implements them over NINA; the harness implements them for offline replay.

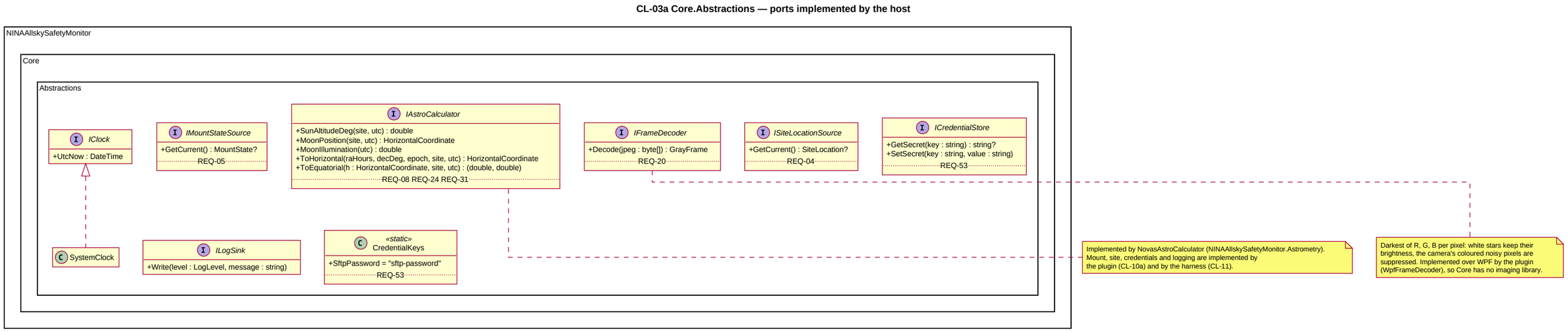


Figure 4-4 CL-03b Settings

The JSON settings tree with its defaults. Initial values match the bracketed defaults in the requirements.

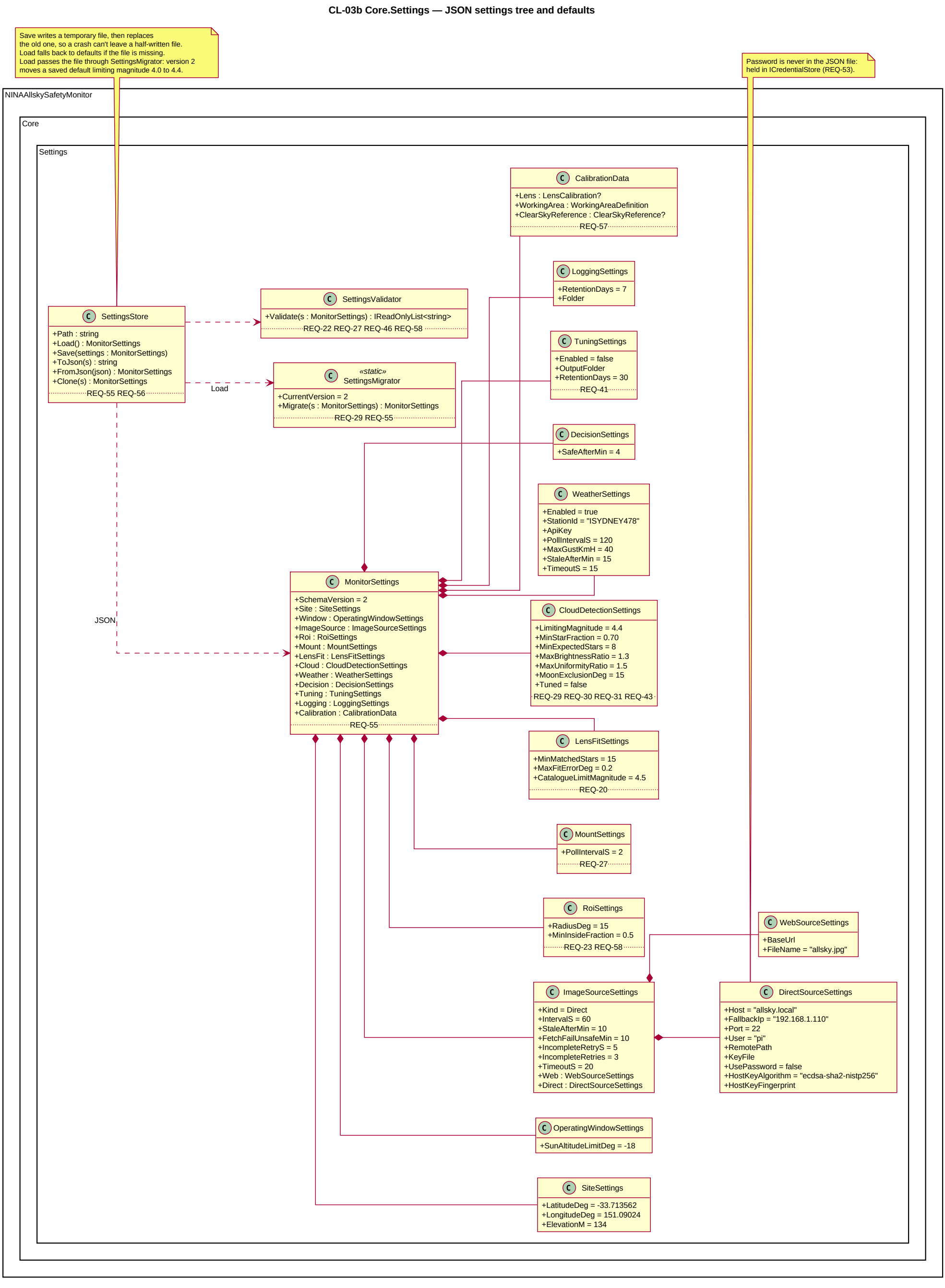


Figure 4-5 CL-04 Image acquisition

Web and SFTP sources behind one interface, SSH.NET kept behind IRemoteFileConnector, host-key type and fingerprint pinning, the incomplete-JPEG retry and the freshness rule.

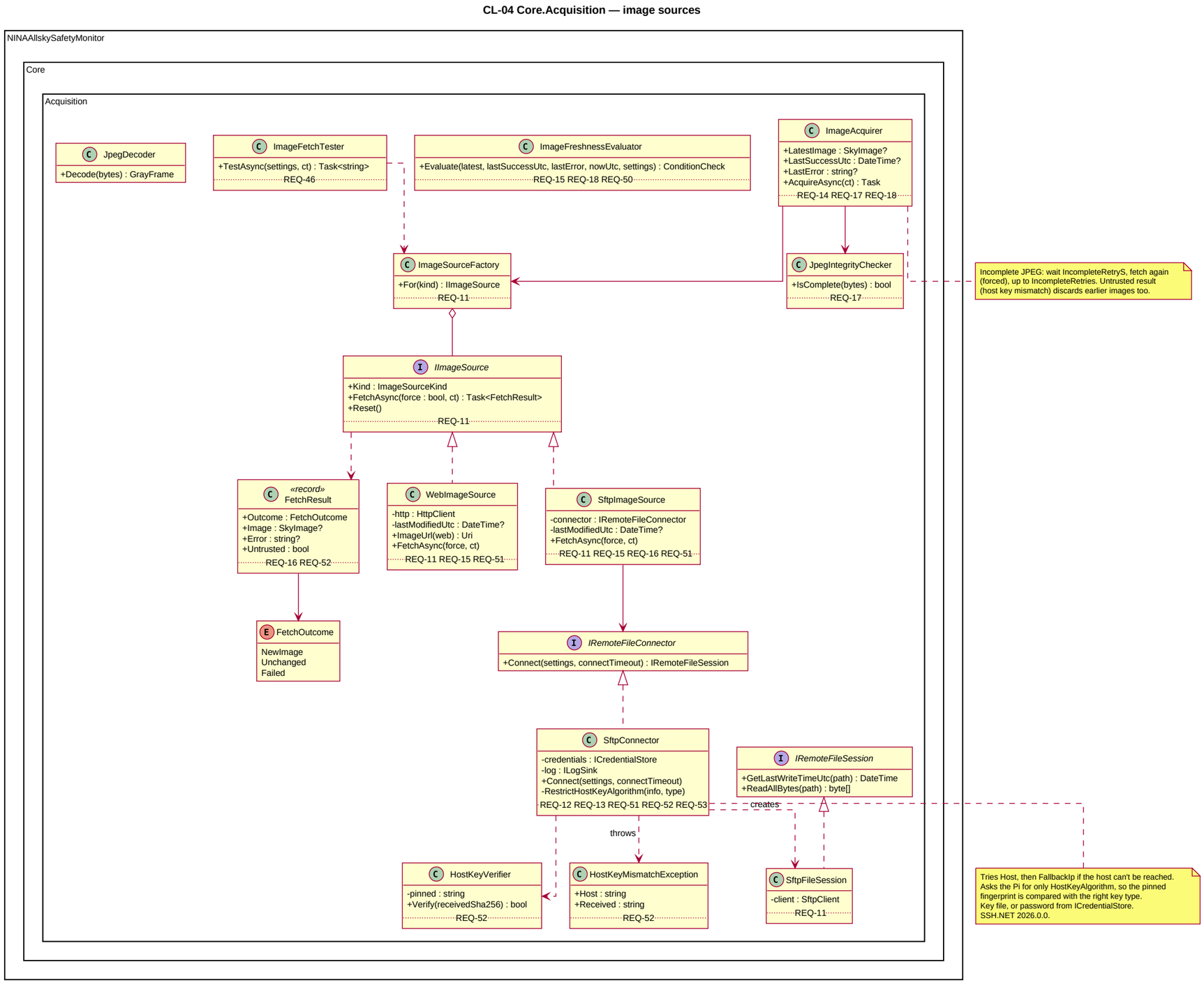


Figure 4-6 CL-05 Weather

PWS client and parser (the API key never appears in messages), a poller asked every cycle that reads the station every 2 minutes, and the evaluator producing the rain, wind and data-age checks.

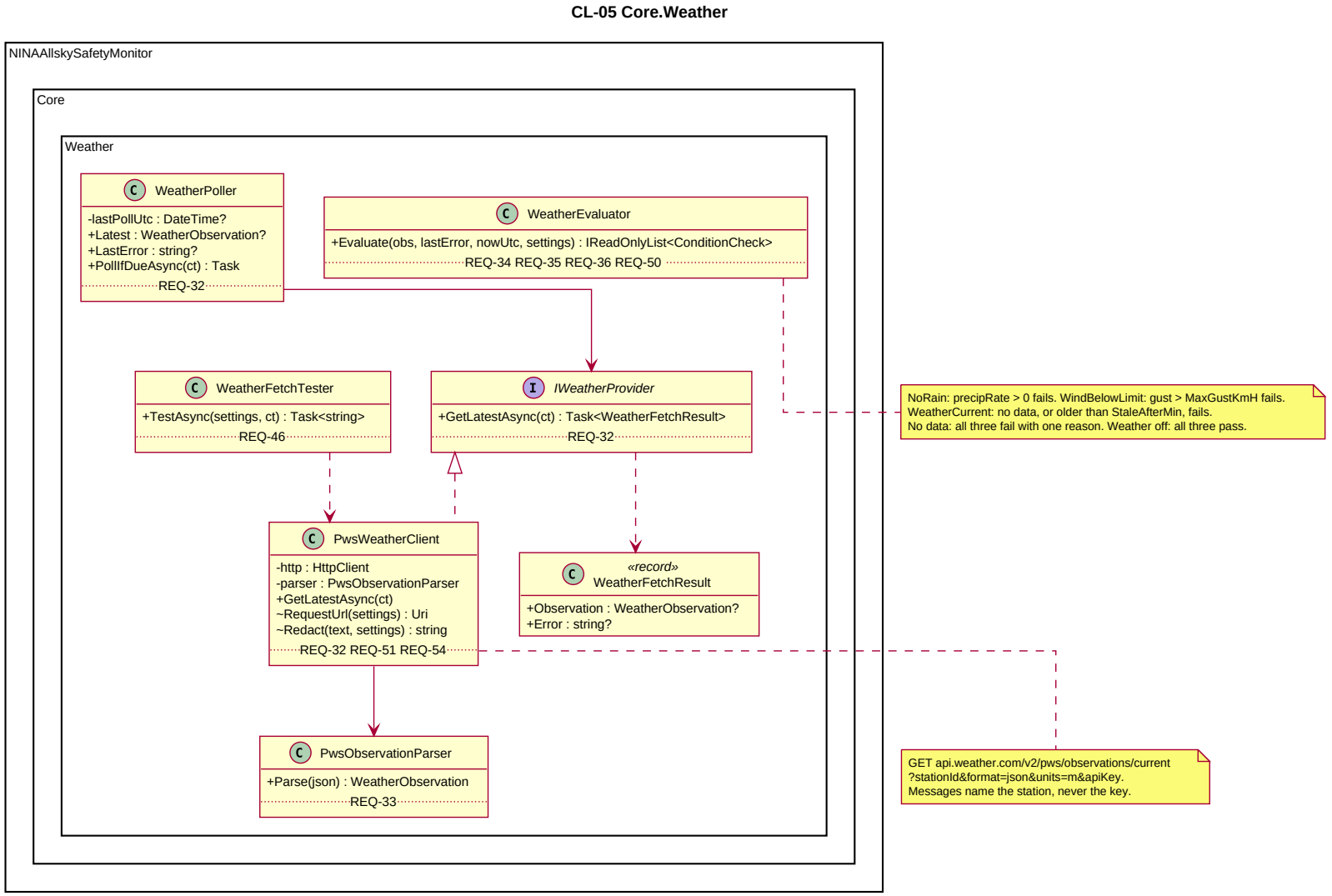


Figure 4-7 CL-06 Geometry

Site resolution, operating window, fisheye lens projection, the working area, and the ROI built from the mount’s RA/Dec and clipped to the working area, or the whole working area as fallback.

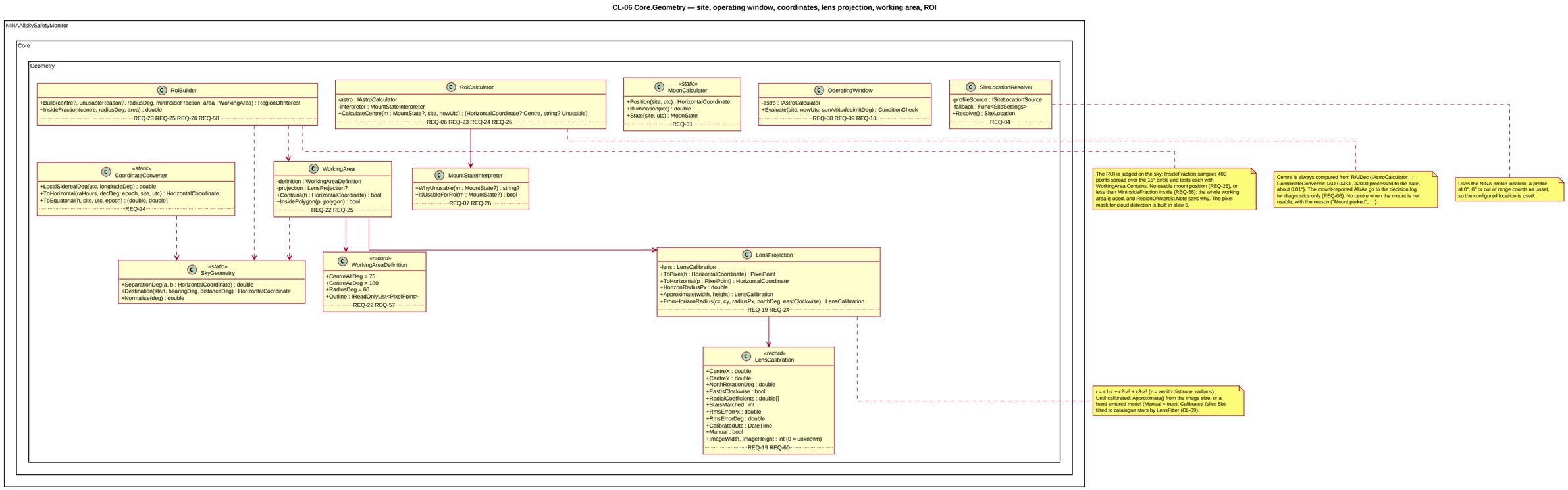


Figure 4-8 CL-07 Sky-frame check and cloud detection

The sky-frame check; catalogue star prediction, detection and matching; the star-fraction and brightness tests against the clear-sky block map; the Moon adjustment and reference capture, combined by CloudDetector.

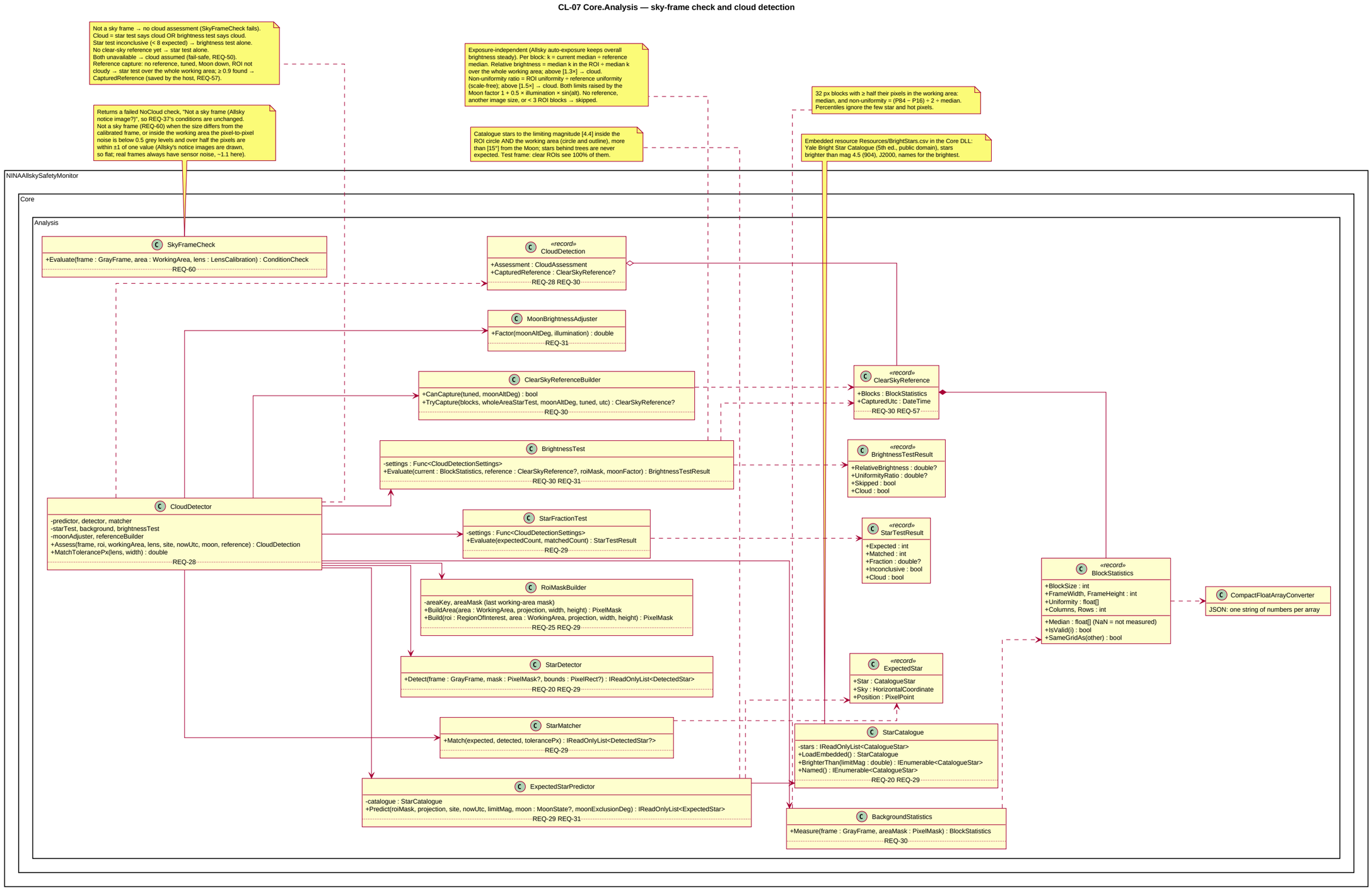


Figure 4-9 CL-08 Decision and monitoring

MonitorService runs each cycle inside FailSafeGuard, then the decision engine (hysteresis, slew hold) and the manual override, then publishes. The builder composes everything and subscribes the decision log.

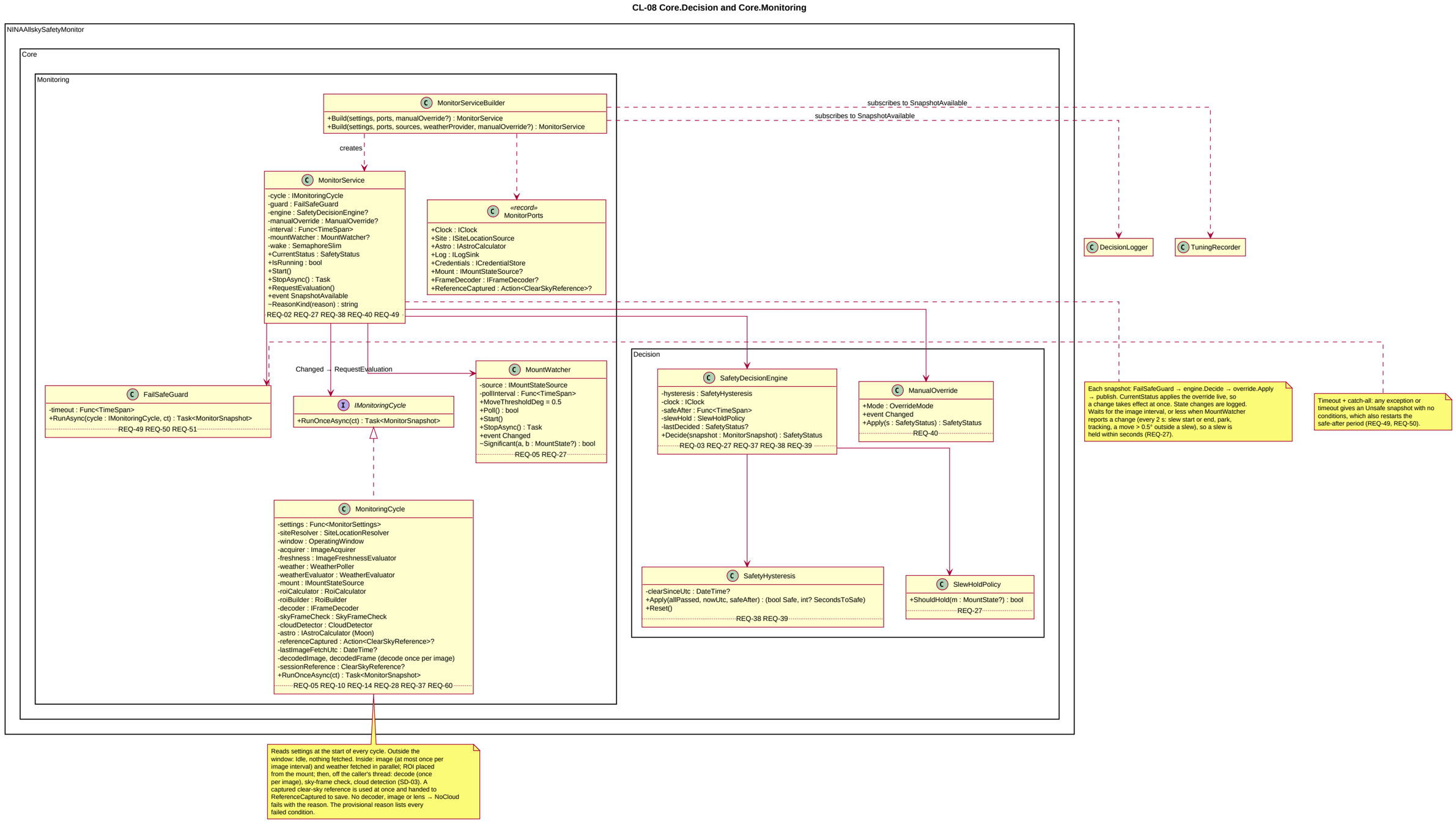


Figure 4-10 CL-09 Calibration, tuning, logging and traceability

Lens calibration from catalogue stars (automatic start, identified stars as fallback, fit-quality warnings), the tuning recorder and replay source, the daily CSV decision log, secret redaction and the Requirement attribute.

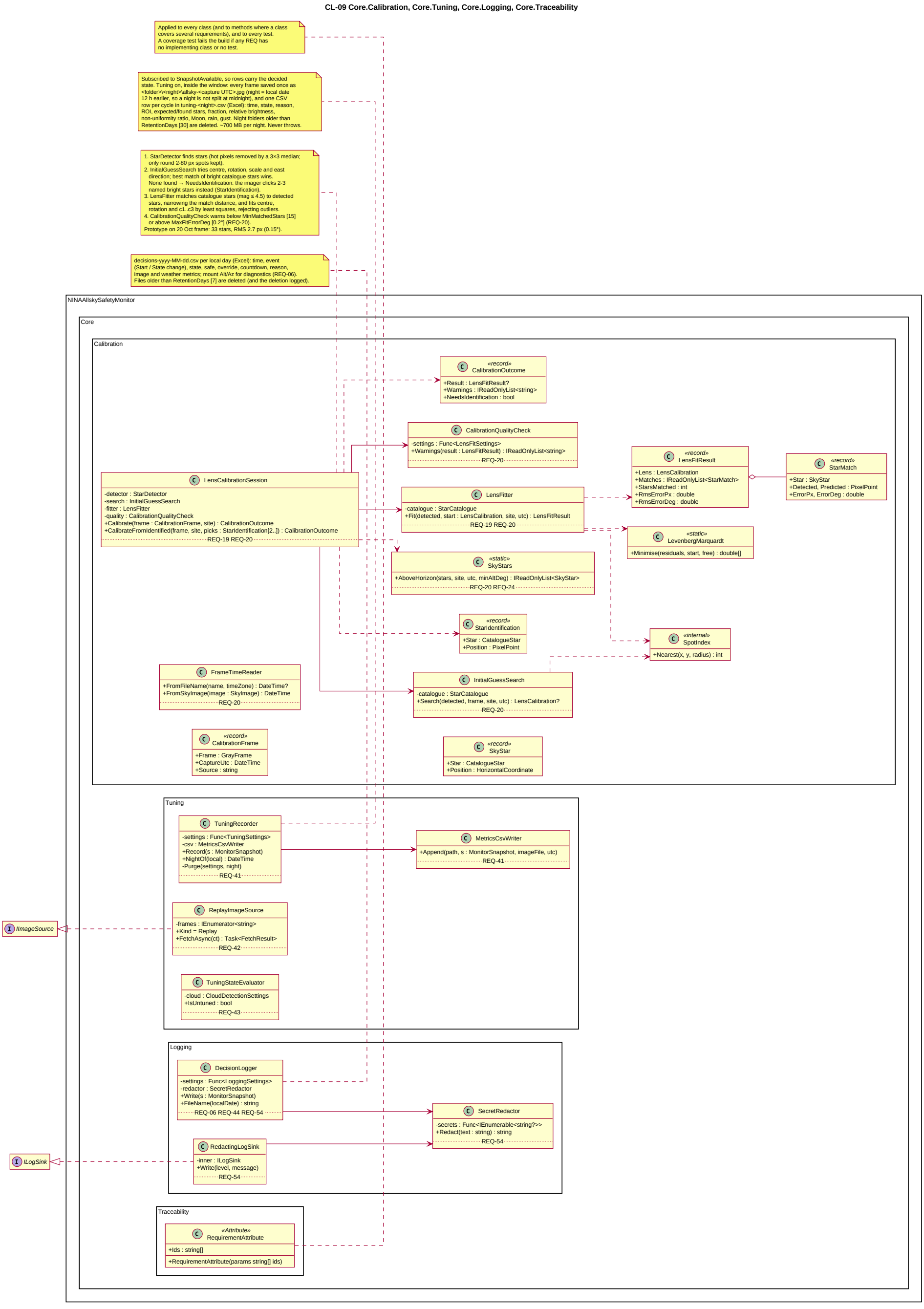


Figure 4-11 CL-10a Plugin: NINA integration

Equipment provider, safety-monitor device, the shared MonitorHost and settings store, and the NINA adapters that implement Core’s ports.

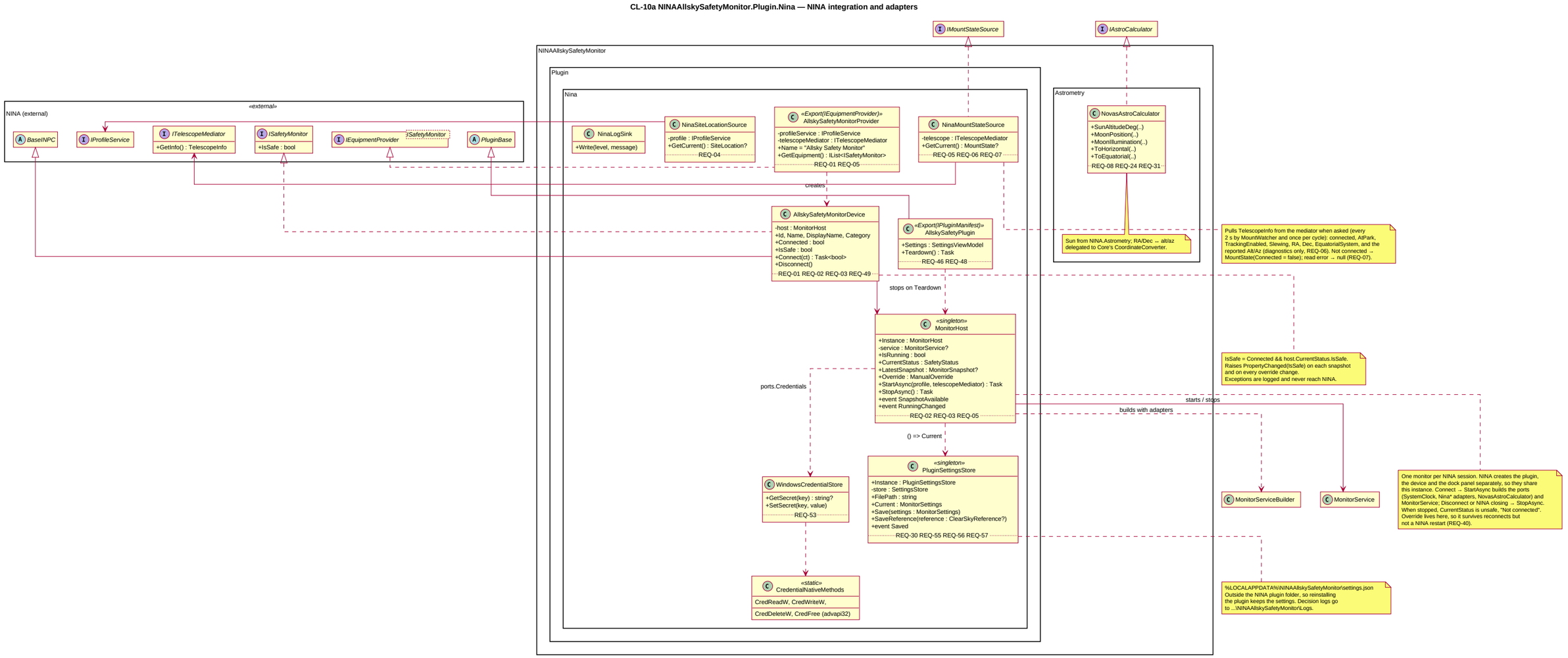


Figure 4-12 CL-10b Plugin: view models and views

MVVM layer inside NINA: the status dock panel, settings page, and the enlarged image viewer with its view and calibration modes.

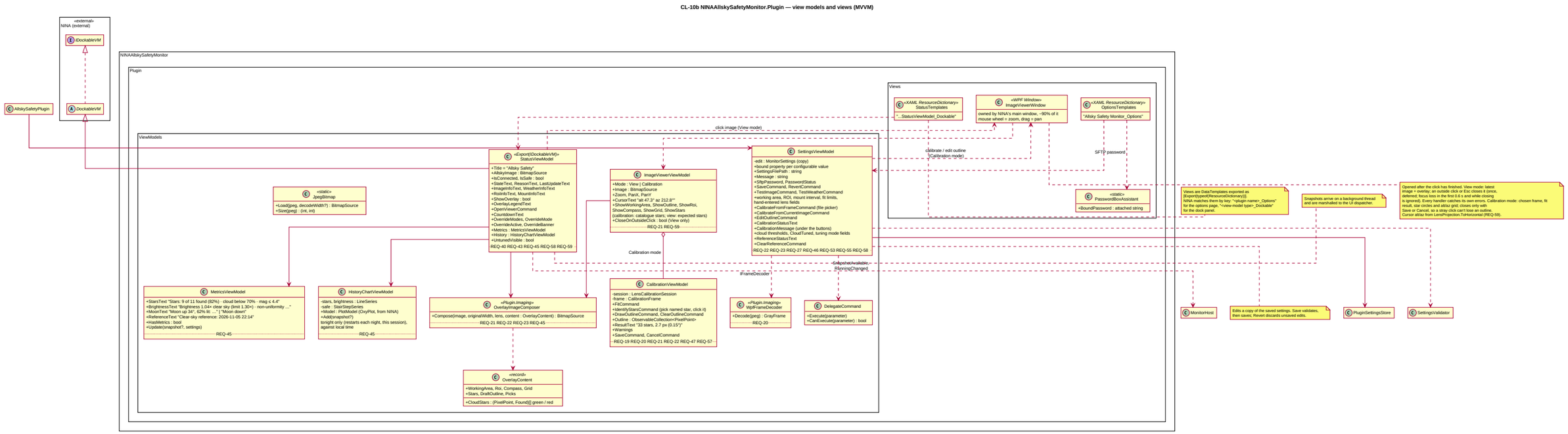
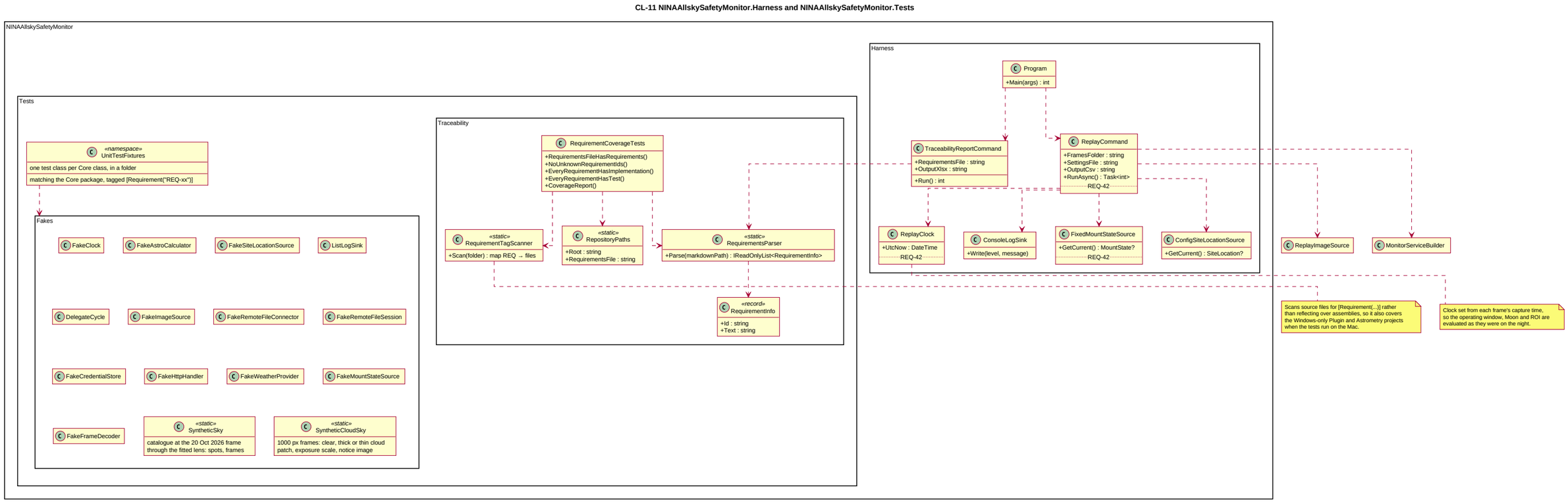


Figure 4-13 CL-11 Harness and tests

Offline replay and traceability-report commands, and the coverage tests.



5 Scenarios

Twelve sequence diagrams cover the main and failure paths. Each title names the use cases and requirements it exercises.

Scenario	Title	Use cases
SD-01	Connect safety monitor	UC-01
SD-02	Monitoring cycle, normal path	UC-04 … UC-10, UC-18
SD-03	Check the frame and detect cloud in the ROI	UC-08
SD-04	Conditions fail and clear	UC-10, UC-03
SD-05	Slewing, fallback and the working area	UC-07
SD-06	Acquire image over SFTP	UC-06
SD-07	Monitor failure and idle window	UC-19, UC-05
SD-08	Calibrate lens and define the working area	UC-12, UC-13
SD-09	Replay saved frames offline	UC-17
SD-10	Manual override and disconnect	UC-11, UC-02
SD-11	View status and edit settings	UC-14, UC-15
SD-12	View the Allsky image enlarged	UC-14

Figure 5-1 SD-01 Connect safety monitor

NINA lists the device from the provider at start-up. Connect asks MonitorHost to start: it registers for mount updates, builds the ports and the monitor and starts its background loop. Each cycle reads the current settings. IsSafe stays false until a safe decision.

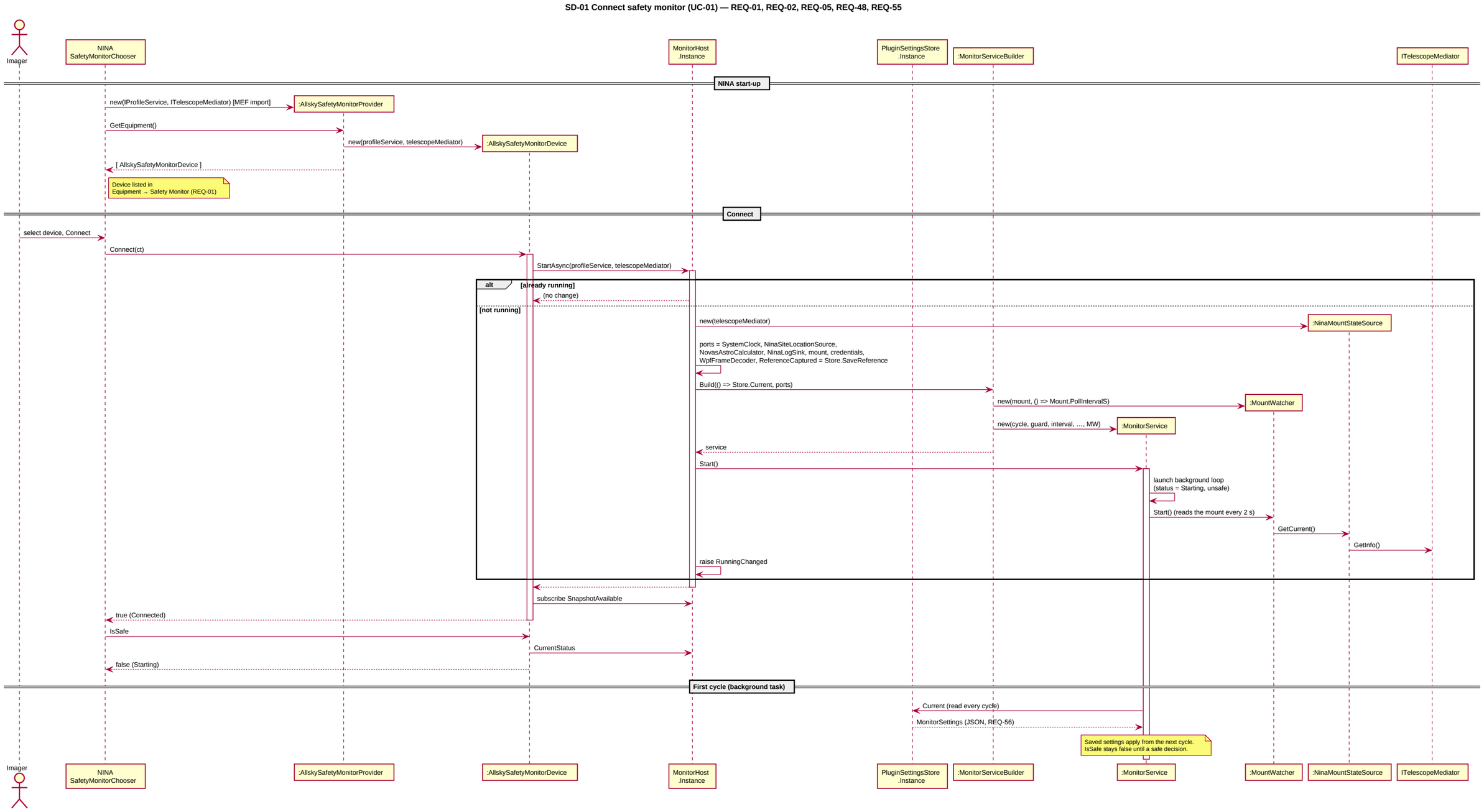


Figure 5-2 SD-02 Monitoring cycle, normal path

One cycle: site and operating window, image and weather in parallel, freshness and weather checks, ROI and cloud; then the decision engine, override, publication and the decision log.

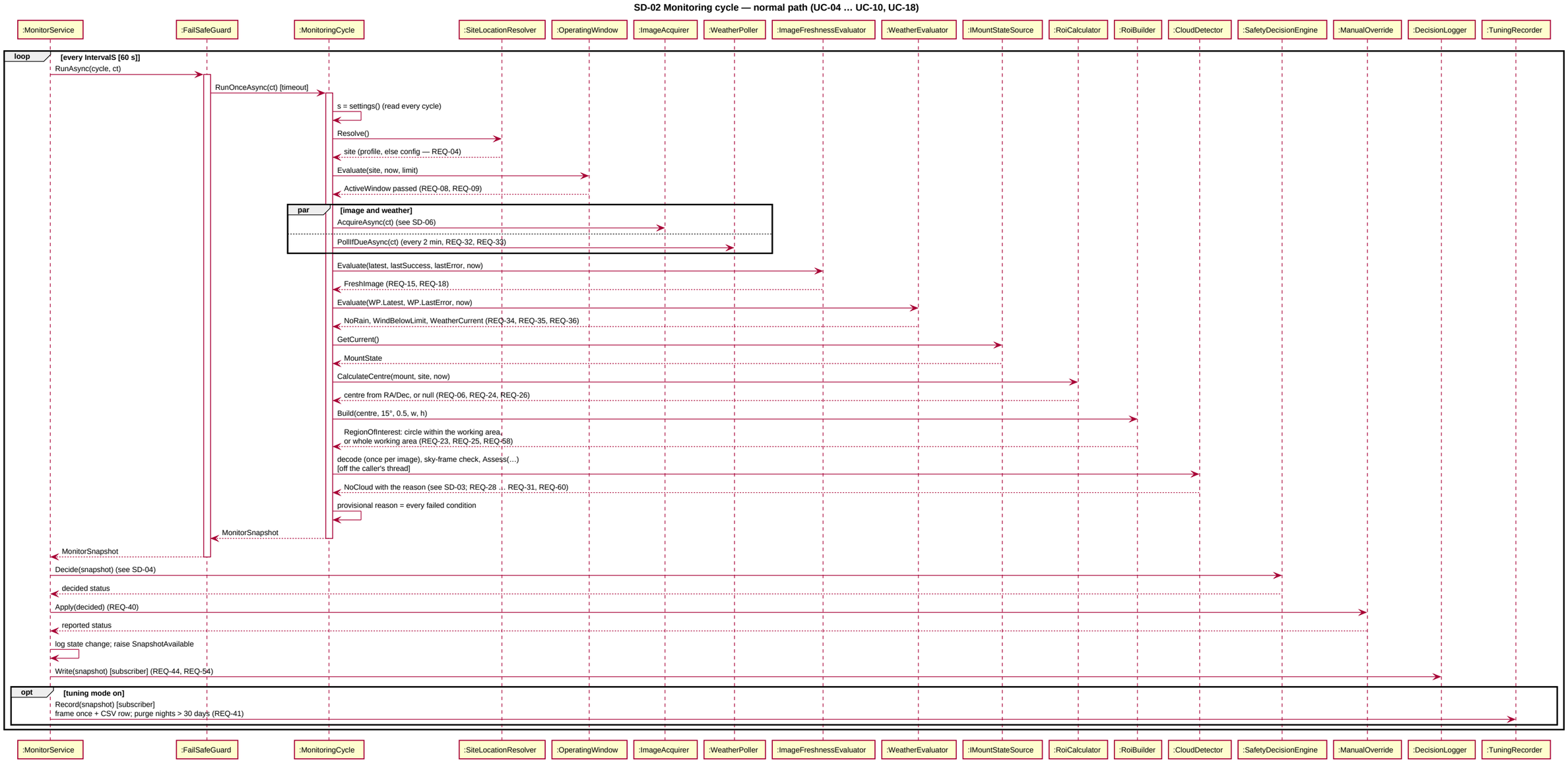


Figure 5-3 SD-03 Check the frame and detect cloud in the ROI

Decode on the darkest channel and reject a frame that is not a sky frame; expected versus detected catalogue stars in the ROI and working area, the inconclusive case below 8 expected stars, the exposure-independent brightness test against the clear-sky map, automatic reference capture, and the fail-safe combination.

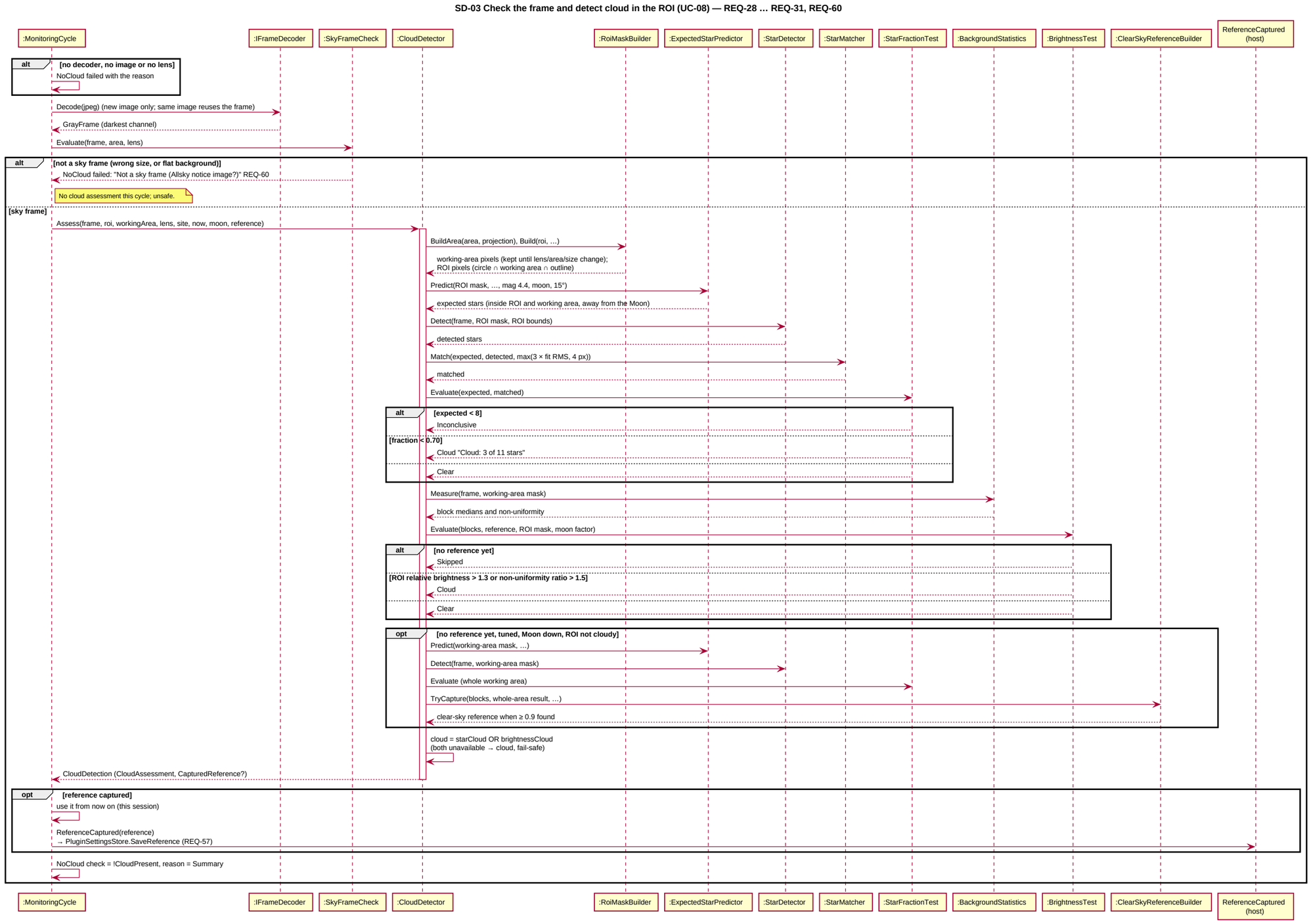


Figure 5-4 SD-04 Conditions fail and clear

Unsafe at once; safe only after four continuous clear minutes, with a countdown. Idle or a failed cycle restarts the period.

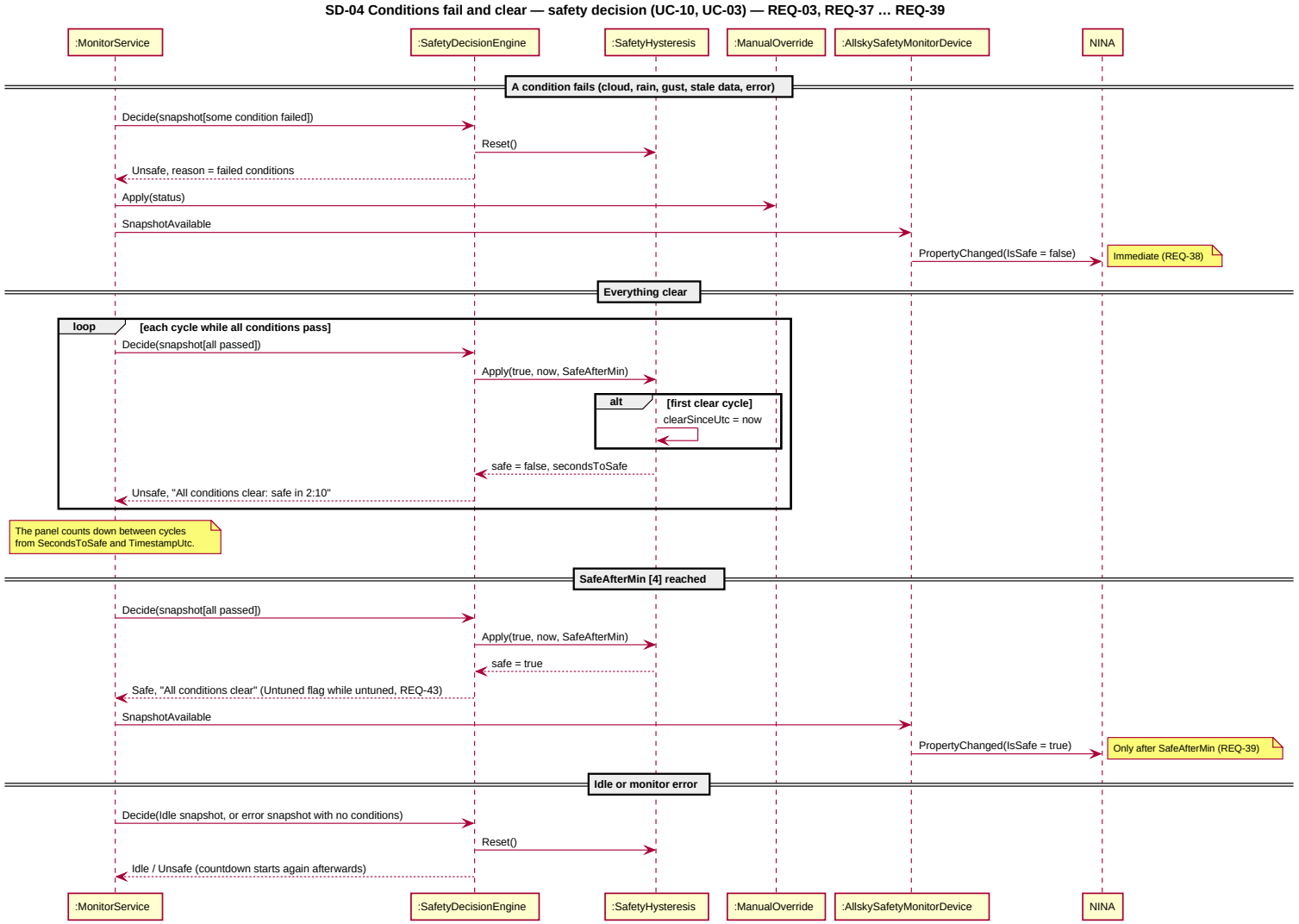


Figure 5-5 SD-05 Slewing, fallback and the working area

MountWatcher reads the mount every 2 s and triggers an immediate re-evaluation on a change. The decision is held while slewing. The ROI is judged against the working area; with no usable mount position, or less than half the ROI inside, the whole working area is judged.

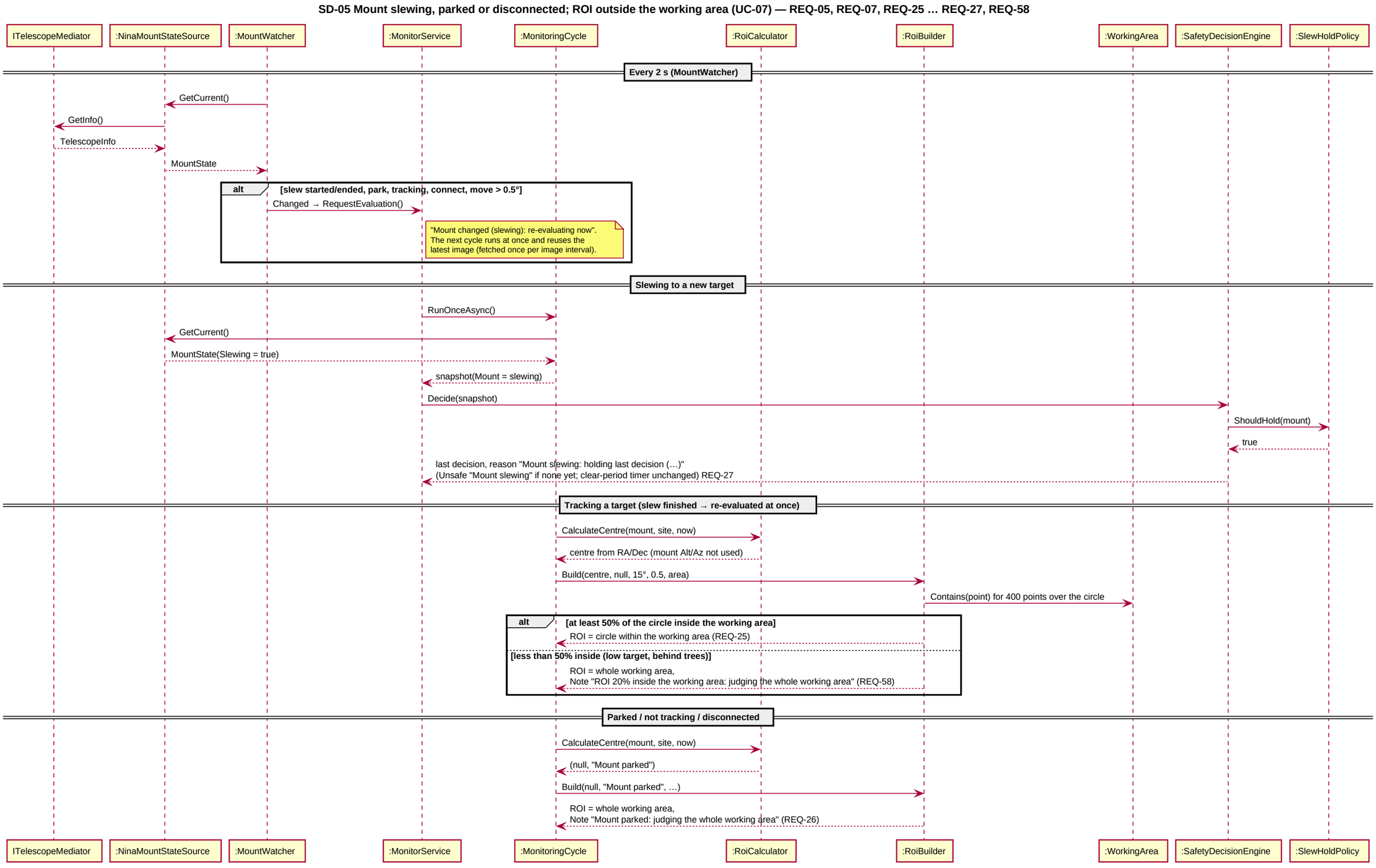


Figure 5-6 SD-06 Acquire image over SFTP

Host-key type restriction and fingerprint check, fallback IP, stat before download, unchanged-image skip, untrusted-image discard and the incomplete-JPEG retry.

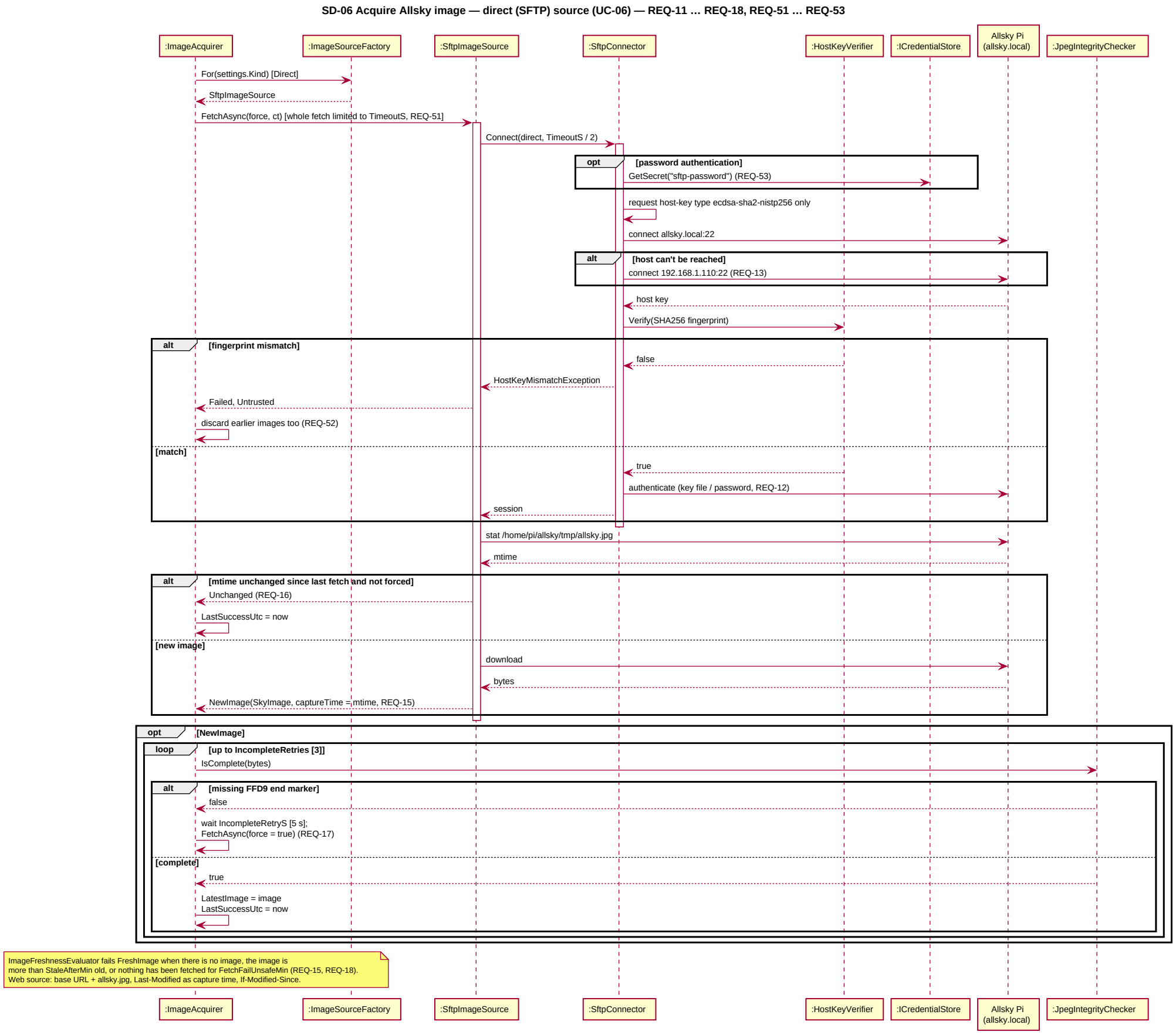


Figure 5-7 SD-07 Monitor failure and idle window

An exception or hang becomes an Unsafe status without reaching NINA. Outside the operating window the cycle idles and reports unsafe.

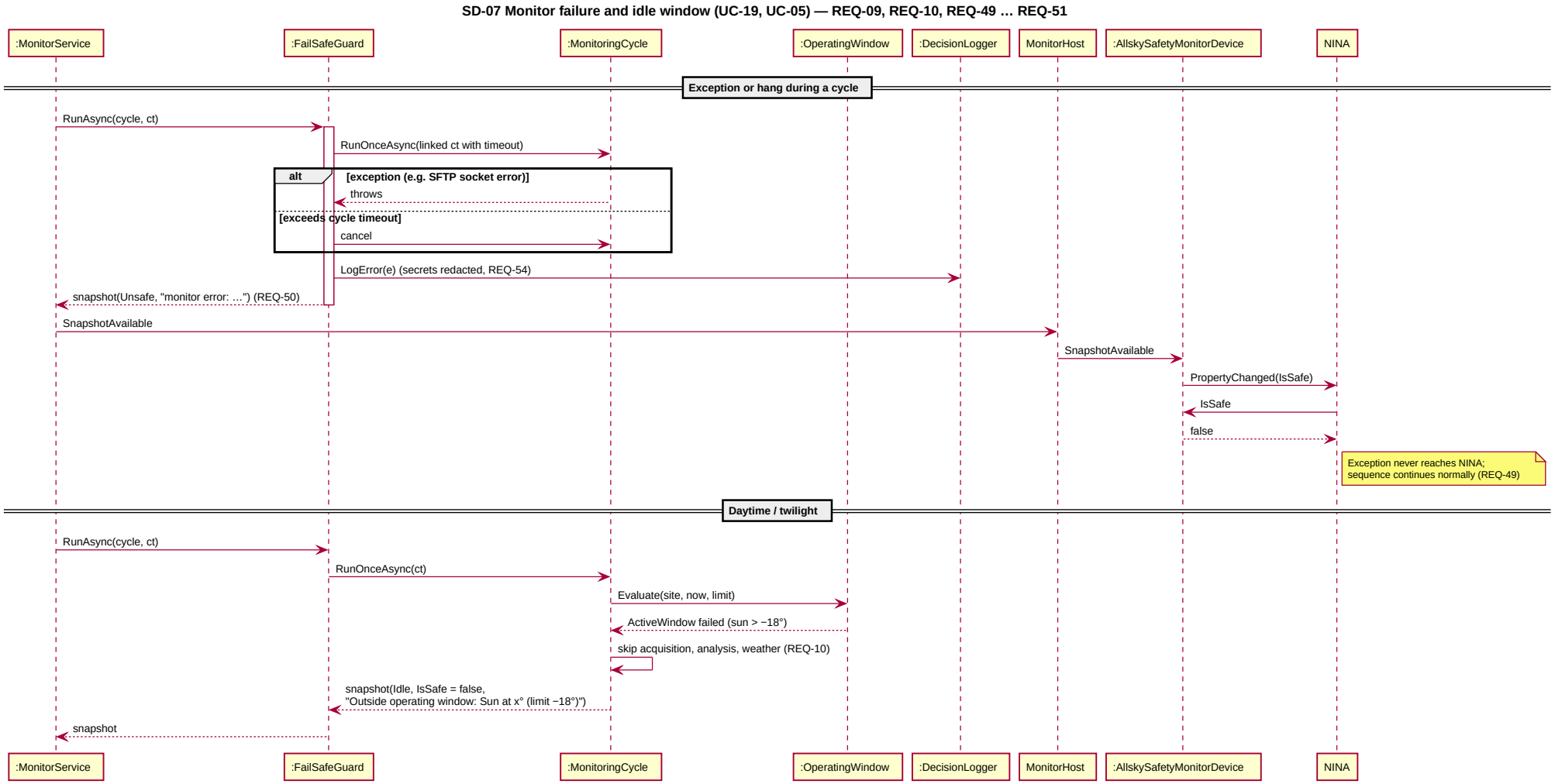


Figure 5-8 SD-08 Calibrate lens and define the working area

From a saved clear frame (time from the file name) or the current image: detect stars, find a starting model automatically or from two or three stars the imager identifies, fit, warn on a poor fit, inspect the overlay, draw the outline, Save.

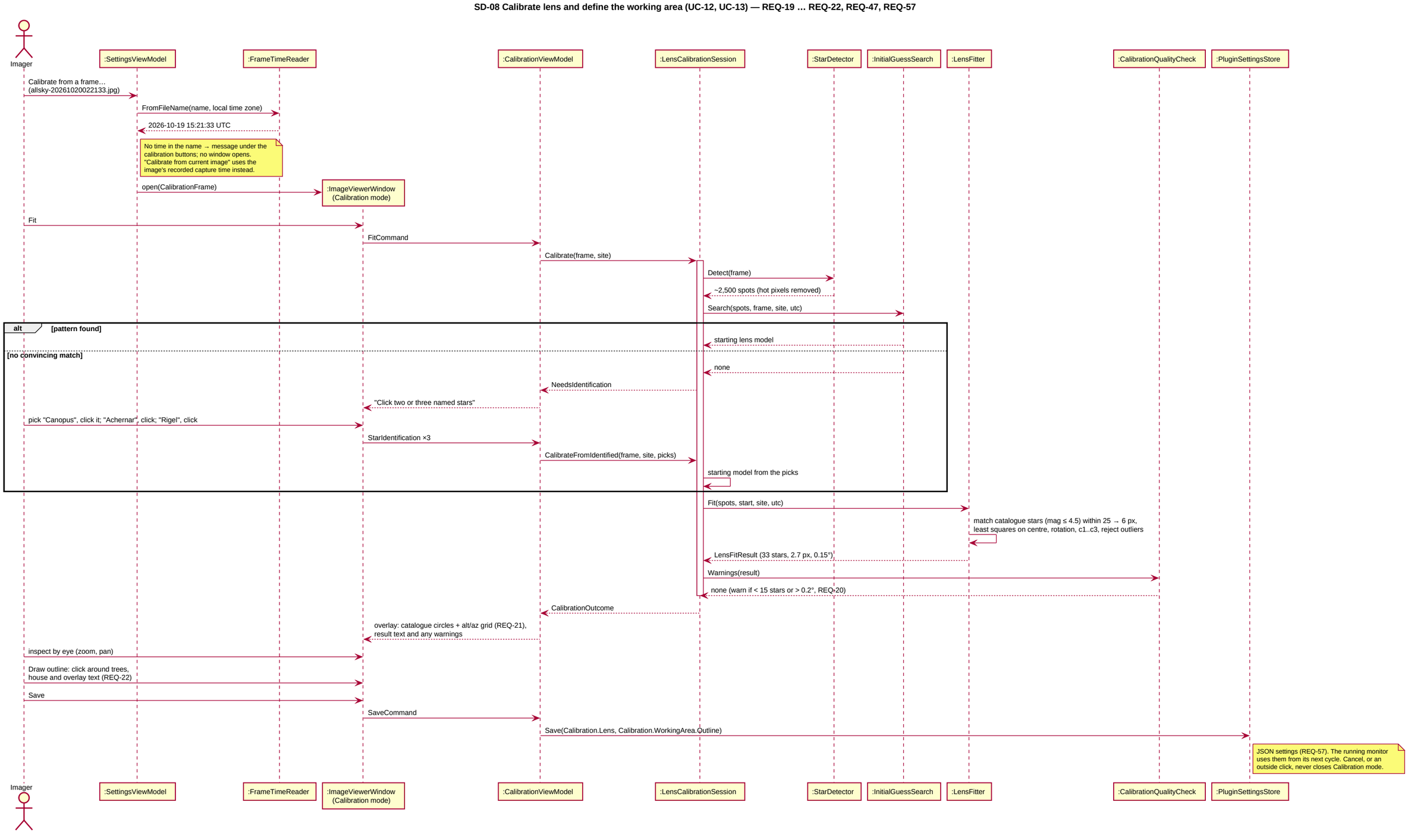


Figure 5-9 SD-09 Replay saved frames offline

Saved frames run through the real cycle with trial settings and a clock set from each frame’s capture time; results go to a CSV for Excel.

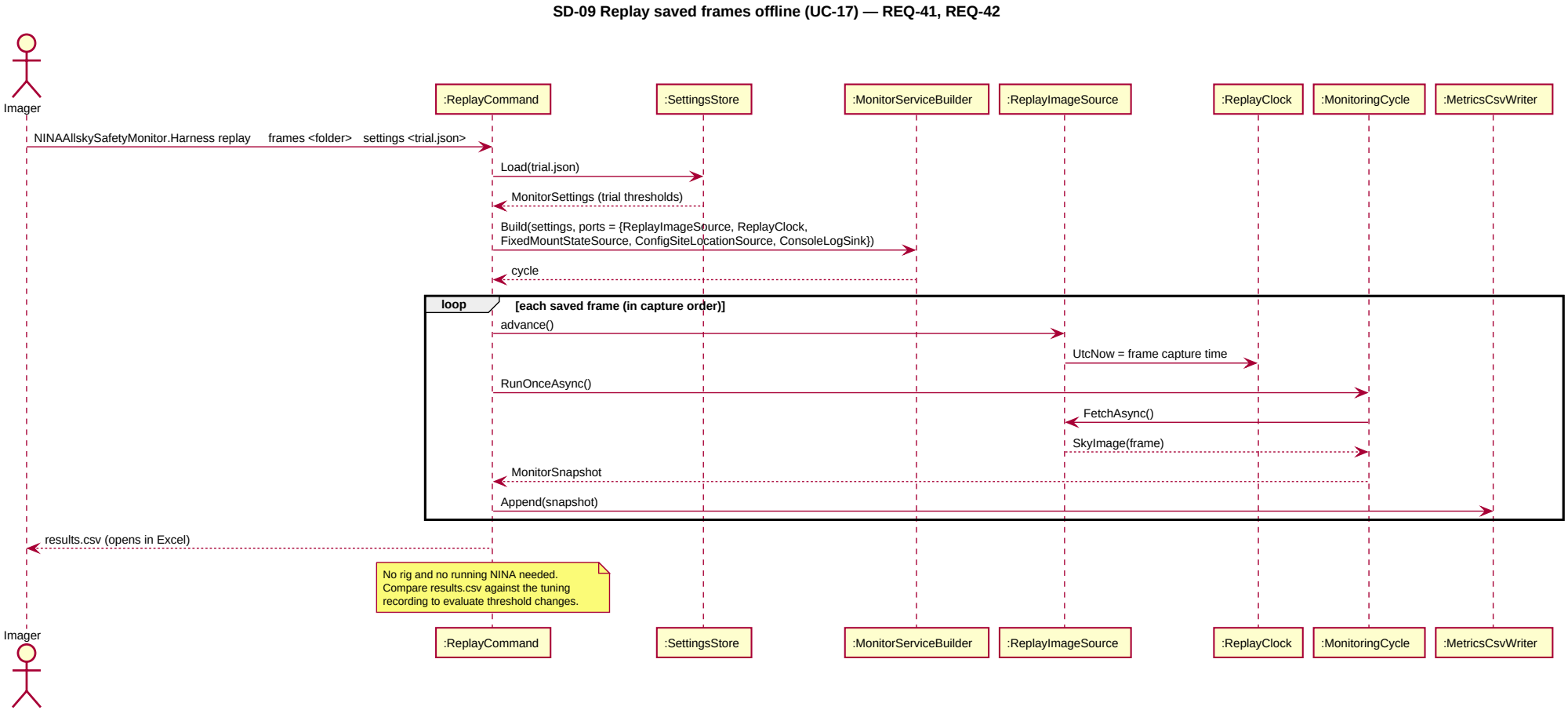


Figure 5-10 SD-10 Manual override and disconnect

The override is held by MonitorHost, logged, applied at once and never saved. Disconnect stops the loop and unregisters from NINA, leaving nothing running.

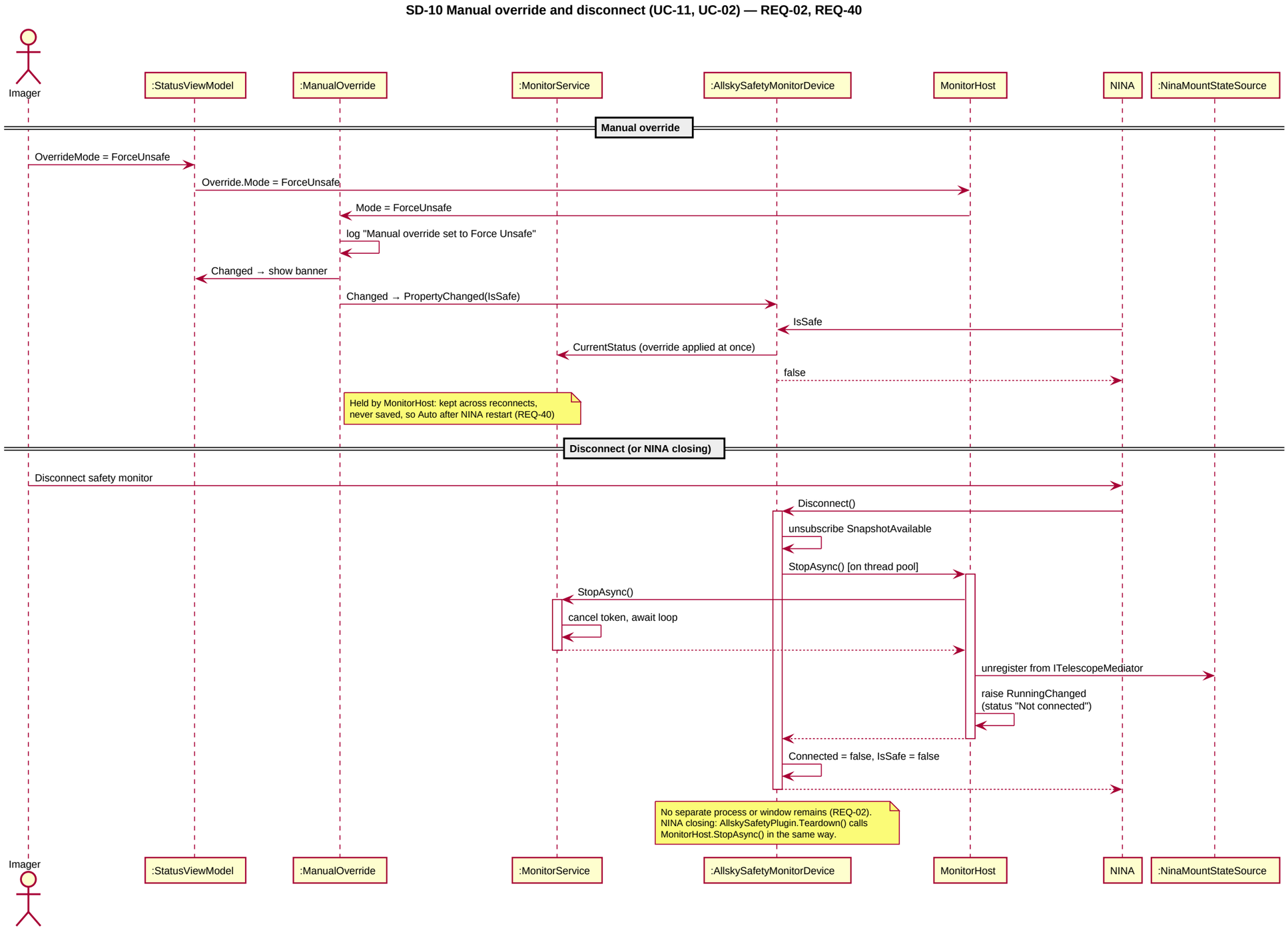


Figure 5-11 SD-11 View status and edit settings

The status panel updates from each snapshot and every 5 s. Settings are validated and can be tested before saving; the password goes to Credential Manager and everything else to JSON.

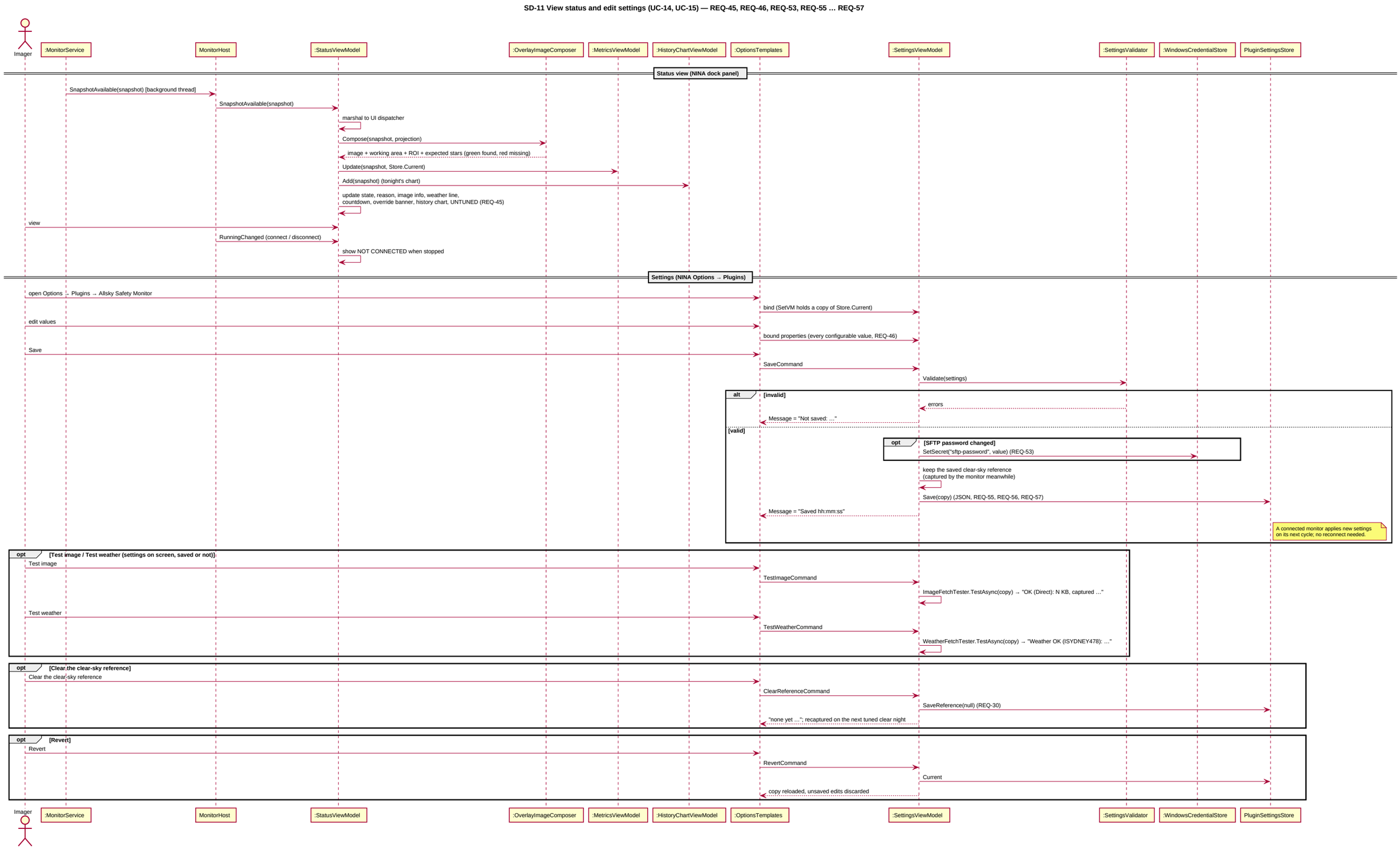
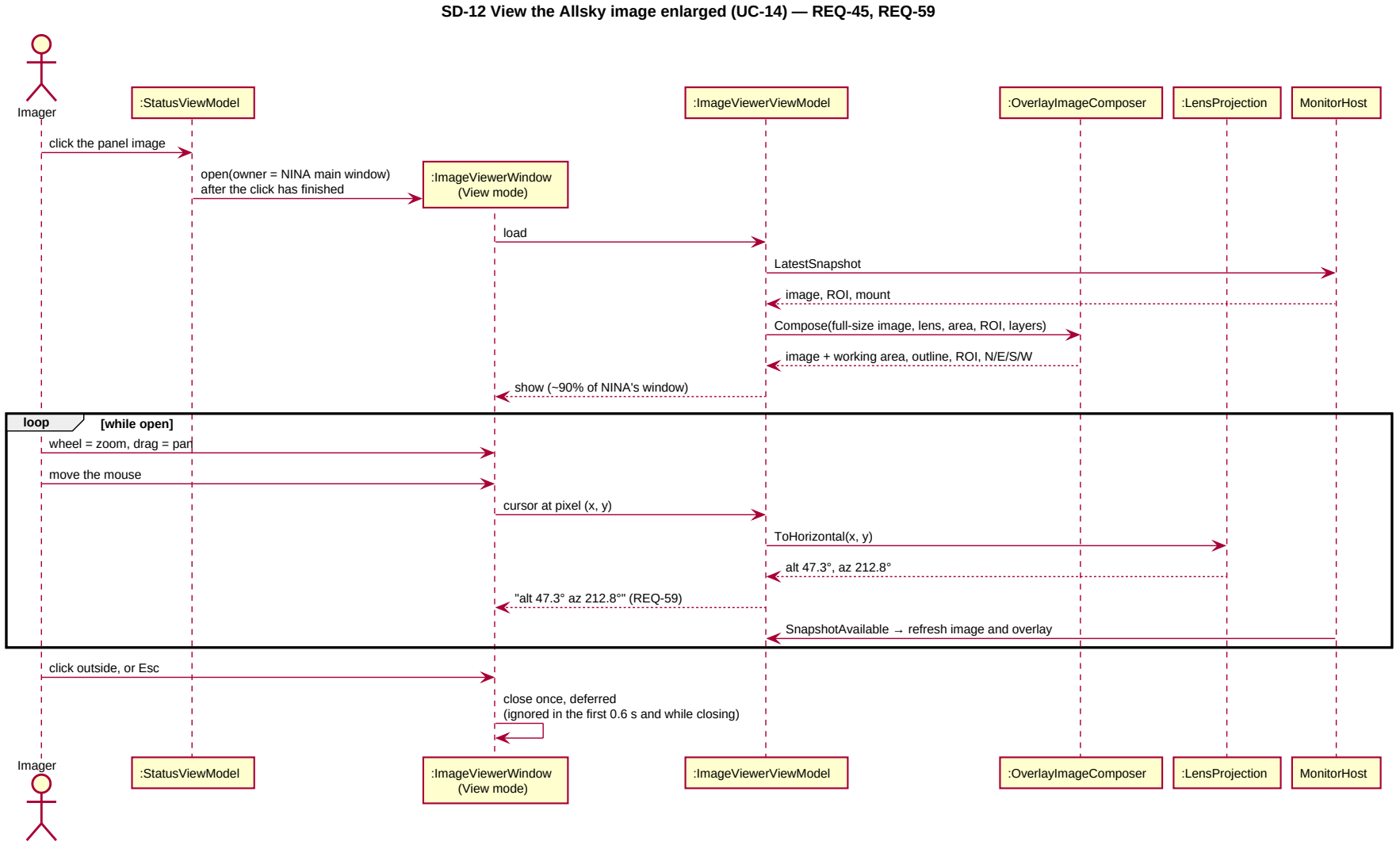


Figure 5-12 SD-12 View the Allsky image enlarged

Clicking the panel image opens a large viewer with the overlay, zoom and pan; the cursor position is converted to altitude and azimuth with the lens model. An outside click or Esc closes it.



6 Traceability matrix

All 60 requirements trace to at least one use case, one design class and one scenario. The matrix is generated from the diagram sources; traceability.xlsx holds the same data with filters.

REQ	Requirement (abridged)	Use cases	Design classes	Scenarios
REQ-01	Appears in NINA as a safety monitor device (Equipment → Safety Monitor), usable by NINA sequenc…	UC-01, UC-02	AllskySafetyMonitorDevice, AllskySafetyMonitorProvider	SD-01
REQ-02	Starts working when the device is connected in NINA and stops when it is disconnected or NINA c…	UC-01, UC-02	AllskySafetyMonitorDevice, MonitorHost, MonitorService	SD-01, SD-10
REQ-03	Reports the result of the safety decision (§9) to NINA as safe or unsafe.	UC-03	AllskySafetyMonitorDevice, MonitorHost, SafetyDecisionEngine, SafetyStatus	SD-04
REQ-04	The site location is taken from the active NINA profile. If that isn't available, a configured …	UC-05, UC-07	ISiteLocationSource, NinaSiteLocationSource, SiteLocation, SiteLocationResolver	SD-02
REQ-05	The mount state is taken from NINA: connected, parked, tracking, slewing, RA (hours), Dec (degr…	UC-07	AllskySafetyMonitorProvider, IMountStateSource, MonitorHost, MonitoringCycle, MountState, MountWatcher, NinaMountStateSource	SD-01, SD-05
REQ-06	The mount-reported Altitude/Azimuth are not used for positioning. The ASI Mount driver reports …	UC-07	DecisionLogger, MountState, NinaMountStateSource, RoiCalculator	SD-02
REQ-07	When the mount reports no valid position (for example, disconnected), the position is treated a…	UC-07	MountState, MountStateInterpreter, NinaMountStateSource	SD-05
REQ-08	Computes sun altitude from the site location and current time.	UC-05	IAstroCalculator, NovasAstroCalculator, OperatingWindow	SD-02
REQ-09	Active only while the sun is below (astronomical twilight).	UC-05	OperatingWindow	SD-02, SD-07
REQ-10	Outside the active window it idles: no image fetching or analysis, and it reports unsafe.	UC-05	MonitoringCycle, OperatingWindow	SD-07
REQ-11	The image source is selectable:	UC-06	IImageSource, ImageSourceFactory, SftpFileSession, SftpImageSource, SkyImage, WebImageSource	SD-06
REQ-12	Direct access authenticates with an SSH key (preferred,) or a password.	UC-06	SftpConnector	SD-06
REQ-13	The Pi host is, with an optional fallback IP in case name resolution fails.	UC-06	SftpConnector	SD-06
REQ-14	Fetches a new image every while active.	UC-06	ImageAcquirer, MonitoringCycle	SD-06
REQ-15	Determines the image capture time (file modification time, or the web server's last-modified ti…	UC-06	ImageFreshnessEvaluator, SftpImageSource, SkyImage, WebImageSource	SD-02, SD-06
REQ-16	For direct access, the file's modification time is checked before downloading, and the image is…	UC-06	FetchResult, SftpImageSource	SD-06

REQ	Requirement (abridged)	Use cases	Design classes	Scenarios
REQ-17	An image caught part-way through being written (an incomplete JPEG) is discarded and fetched ag…	UC-06	ImageAcquirer, JpegIntegrityChecker	SD-06
REQ-18	If no image can be fetched for, reports unsafe.	UC-06	ImageAcquirer, ImageFreshnessEvaluator	SD-02, SD-06
REQ-60	An image that is not a sky frame, such as the notice image Allsky shows when its software is no…	UC-06	LensCalibration, MonitoringCycle, SkyFrameCheck	SD-02, SD-03
REQ-19	Provides a one-off calibration that maps alt/az to Allsky image pixel positions: image centre, …	UC-12	CalibrationViewModel, LensCalibration, LensCalibrationSession, LensFitter, LensProjection	SD-08
REQ-20	Calibration is done by matching identified catalogue stars in a real clear-sky frame. A calibra…	UC-12	CalibrationQualityCheck, CalibrationViewModel, FrameTimeReader, IFrameDecoder, InitialGuessSearch, LensCalibrationSession, LensFitSettings, LensFitter, SkyStars, StarCatalogue, StarDetector, WpfFrameDecoder	SD-08
REQ-21	Shows a verification overlay (catalogue stars and an alt/az grid drawn on the live image) so ca…	UC-12	CalibrationViewModel, ImageViewerViewModel, OverlayImageComposer	SD-08
REQ-22	A working area defines the part of the sky the monitor can see and assess. It is a circle of co…	UC-13	CalibrationViewModel, OverlayImageComposer, SettingsValidator, SettingsViewModel, WorkingArea, WorkingAreaDefinition	SD-08
REQ-23	The ROI is a circle of radius on the sky, centred on the mount's current RA/Dec.	UC-07	OverlayImageComposer, RegionOfInterest, RoiBuilder, RoiCalculator, RoiSettings, SettingsViewModel	SD-02
REQ-24	RA/Dec is converted to alt/az using the coordinate epoch (JNOW on this rig), the site location …	UC-07	CoordinateConverter, IAstroCalculator, LensProjection, NovasAstroCalculator, RoiCalculator, SkyStars	SD-02
REQ-25	Any part of the ROI outside the working area (REQ-22) is excluded from analysis.	UC-07	RegionOfInterest, RoiBuilder, RoiMaskBuilder, WorkingArea	SD-02, SD-05
REQ-58	If less than a configurable fraction of the ROI lies inside the working area, the whole working…	UC-07	RegionOfInterest, RoiBuilder, RoiSettings, SettingsValidator, SettingsViewModel, StatusViewModel	SD-02, SD-05
REQ-26	The ROI falls back to the whole working area when the mount is not connected, is parked, is not…	UC-07	MountStateInterpreter, RegionOfInterest, RoiBuilder, RoiCalculator	SD-02, SD-05
REQ-27	While the mount is slewing, the current safety state is held and the ROI is updated once the sl…	UC-07	MonitorService, MountSettings, MountWatcher, SafetyDecisionEngine, SettingsValidator, SettingsViewModel, SlewHoldPolicy	SD-05
REQ-28	Any cloud in the ROI means unsafe.	UC-08	CloudAssessment, CloudDetection, CloudDetector, MonitoringCycle	SD-02, SD-03
REQ-29	Primary test: predict which catalogue stars should be visible in the ROI (down to a limiting ma…	UC-08	CloudDetectionSettings, ExpectedStarPredictor, RoiMaskBuilder, SettingsMigrator, StarCatalogue, StarDetector, StarFractionTest, StarMatcher	SD-03
REQ-30	Secondary test, comparing the ROI with a clear-sky reference: a map of sky brightness and non-u…	UC-08	BackgroundStatistics, BrightnessTest, ClearSkyReference, ClearSkyReferenceBuilder, CloudDetection, CloudDetectionSettings, PluginSettingsStore	SD-03, SD-11

REQ	Requirement (abridged)	Use cases	Design classes	Scenarios
REQ-31	Stars within of the Moon are excluded from the expected list, and brightness thresholds are adj…	UC-08	BrightnessTest, CloudDetectionSettings, ExpectedStarPredictor, IAstroCalculator, MoonBrightnessAdjuster, MoonCalculator, MoonState, NovasAstroCalculator	SD-02, SD-03
REQ-32	Polls the Weather.com PWS current-observations API for station every, with metric units, authen…	UC-09	IWeatherProvider, PwsWeatherClient, WeatherPoller	SD-02
REQ-33	Values are read from observations: epoch (observation time), metric.precipRate (mm/h), metric.w…	UC-09	PwsObservationParser, WeatherObservation	SD-02
REQ-34	Any precipitation rate above zero → unsafe immediately.	UC-09	WeatherEvaluator	SD-02
REQ-35	Wind gust above → unsafe.	UC-09	WeatherEvaluator	SD-02
REQ-36	If the station data is older than or unavailable → unsafe. Weather checking can be switched off…	UC-09	WeatherEvaluator	SD-02
REQ-37	The overall status is safe only when all of these are true: active window, fresh image, no clou…	UC-10	MonitoringCycle, SafetyConditions, SafetyDecisionEngine	SD-04
REQ-38	Transition to unsafe is immediate when any condition fails.	UC-10	MonitorService, SafetyDecisionEngine, SafetyHysteresis	SD-04
REQ-39	Transition to safe happens only after all conditions have held continuously for.	UC-10	SafetyDecisionEngine, SafetyHysteresis, SafetyStatus	SD-04
REQ-40	A manual override (Force Safe / Force Unsafe / Auto) is provided for testing, shown prominently…	UC-11	ManualOverride, MonitorService, SafetyStatus, StatusViewModel	SD-02, SD-10
REQ-41	A tuning mode saves every frame together with its computed metrics, in a format readable by Exc…	UC-16	MetricsCsvWriter, MonitorSnapshot, TuningRecorder, TuningSettings	SD-02, SD-09
REQ-42	Saved frames can be re-analysed offline, without the rig or a live NINA session, to test thresh…	UC-17	FixedMountStateSource, ReplayClock, ReplayCommand, ReplayImageSource	SD-09
REQ-43	Until the thresholds have been tuned, conservative defaults (biased towards unsafe) are used an…	UC-16	CloudDetectionSettings, SafetyStatus, StatusViewModel, TuningStateEvaluator	SD-04
REQ-44	Keeps a rolling log of state changes, metrics, errors and the reason for every safe/unsafe deci…	UC-18	DecisionLogger	SD-02
REQ-45	A status view shows:	UC-14	CloudAssessment, HistoryChartViewModel, MetricsViewModel, MonitorSnapshot, OverlayImageComposer, StatusViewModel	SD-11, SD-12
REQ-46	A settings view edits every configurable value.	UC-15	AllskySafetyPlugin, ImageFetchTester, SettingsValidator, SettingsViewModel, WeatherFetchTester	SD-11
REQ-47	A calibration view supports lens calibration (REQ-19–21) and defining the working area (REQ-22)…	UC-12, UC-13	CalibrationViewModel	SD-08
REQ-59	The Allsky image, with its overlay, can be viewed enlarged, and the altitude and azimuth of any…	UC-14	ImageViewerViewModel, StatusViewModel	SD-12

REQ	Requirement (abridged)	Use cases	Design classes	Scenarios
REQ-48	Runs on the Windows AstroPC within NINA 3.x (currently the 3.3 nightly).	UC-01	AllskySafetyPlugin	SD-01
REQ-49	A failure in the safety monitor (crash, hang, network fault) must not disrupt NINA or a running…	UC-19	AllskySafetyMonitorDevice, FailSafeGuard, MonitorService	SD-07
REQ-50	Fail-safe: any failure, or any state that can't be determined, results in unsafe.	UC-19	FailSafeGuard, ImageFreshnessEvaluator, WeatherEvaluator	SD-07
REQ-51	Every network operation (image fetch, weather poll) has a timeout, so a stalled connection can'…	UC-19	FailSafeGuard, PwsWeatherClient, SftpConnector, SftpImageSource, WebImageSource	SD-06, SD-07
REQ-52	The Pi's host key is verified against a pinned fingerprint; a mismatch is reported and the imag…	UC-06	FetchResult, HostKeyMismatchException, HostKeyVerifier, SftpConnector	SD-06
REQ-53	An SFTP password, if used, is stored using Windows' credential storage, never in the settings f…	UC-06, UC-15	CredentialKeys, ICredentialStore, SettingsViewModel, SftpConnector, WindowsCredentialStore	SD-06, SD-11
REQ-54	Passwords and API keys are never written to the log.	UC-18	DecisionLogger, PwsWeatherClient, RedactingLogSink, SecretRedactor	SD-02, SD-07
REQ-55	All configurable values persist between sessions.	UC-15	MonitorSettings, PluginSettingsStore, SettingsMigrator, SettingsStore, SettingsViewModel	SD-01, SD-11
REQ-56	Settings are stored in a JSON file.	UC-15	PluginSettingsStore, SettingsStore	SD-01, SD-11
REQ-57	Calibration results (REQ-19), the working area (REQ-22) and the clear-sky reference (REQ-30) ar…	UC-15	CalibrationData, CalibrationViewModel, ClearSkyReference, PluginSettingsStore, WorkingAreaDefinition	SD-03, SD-08, SD-11